

BFS Traversal

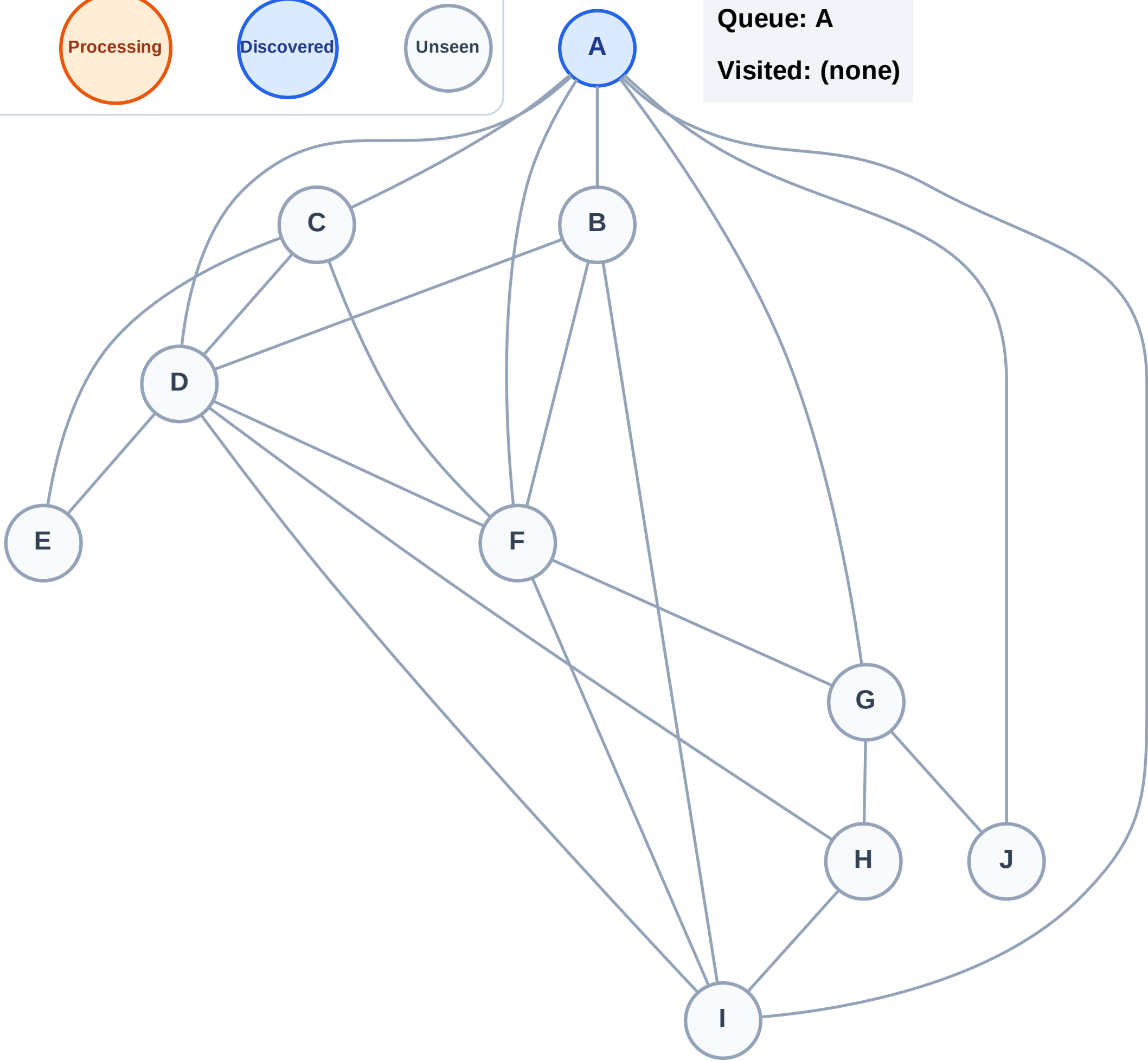
Step 1: Start at A

Legend



Queue: A

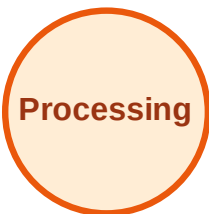
Visited: (none)



BFS Traversal

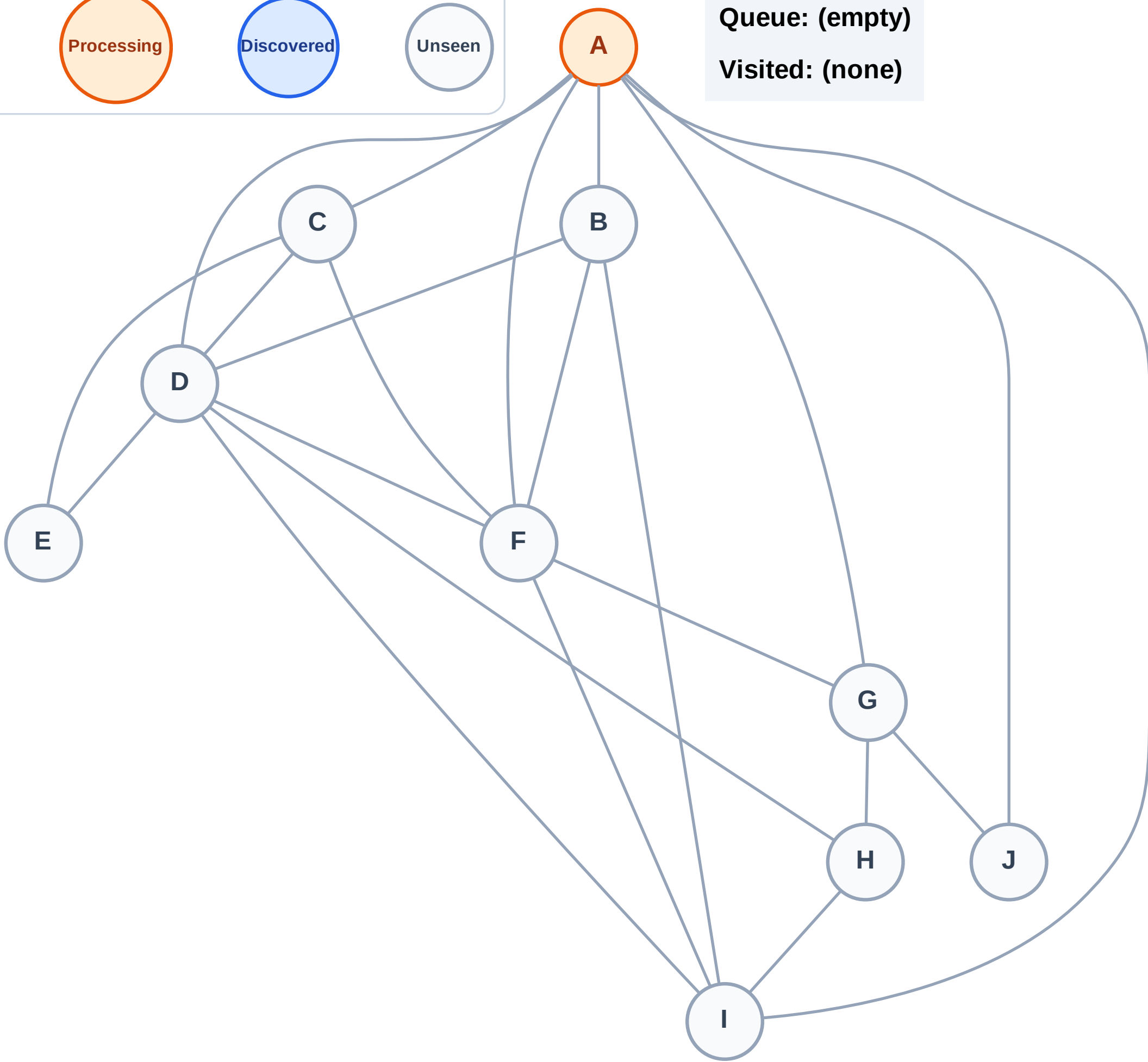
Step 2: Process node A

Legend



Queue: (empty)

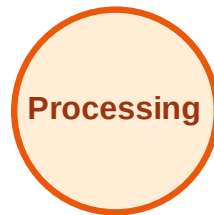
Visited: (none)



BFS Traversal

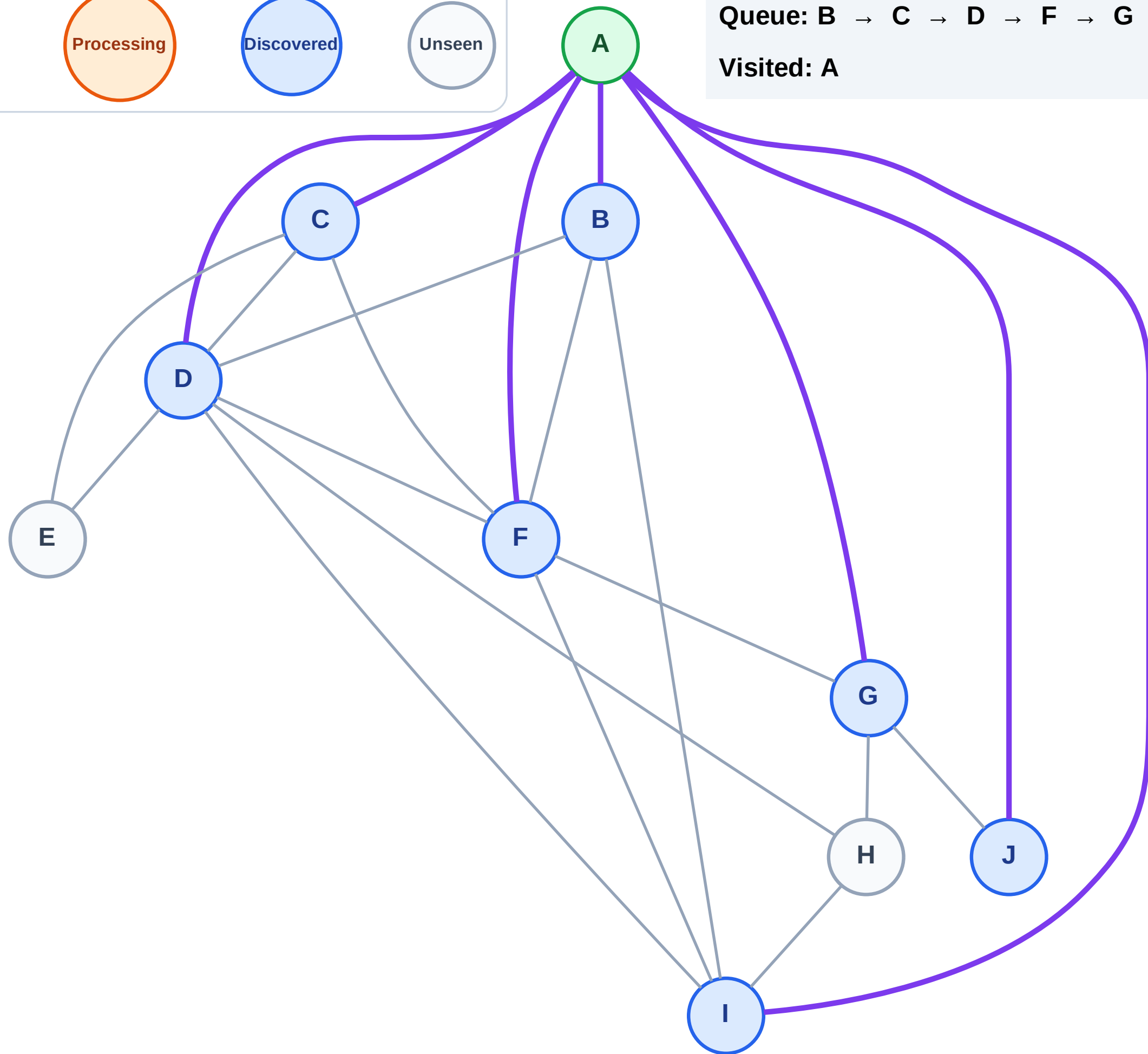
Step 3: Discover B, C, D, F, G, I, J from A

Legend



Queue: B → C → D → F → G → I → J

Visited: A



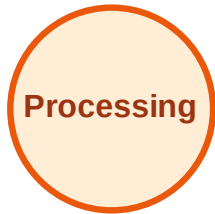
BFS Traversal

Step 4: Process node B

Legend



Complete



Processing



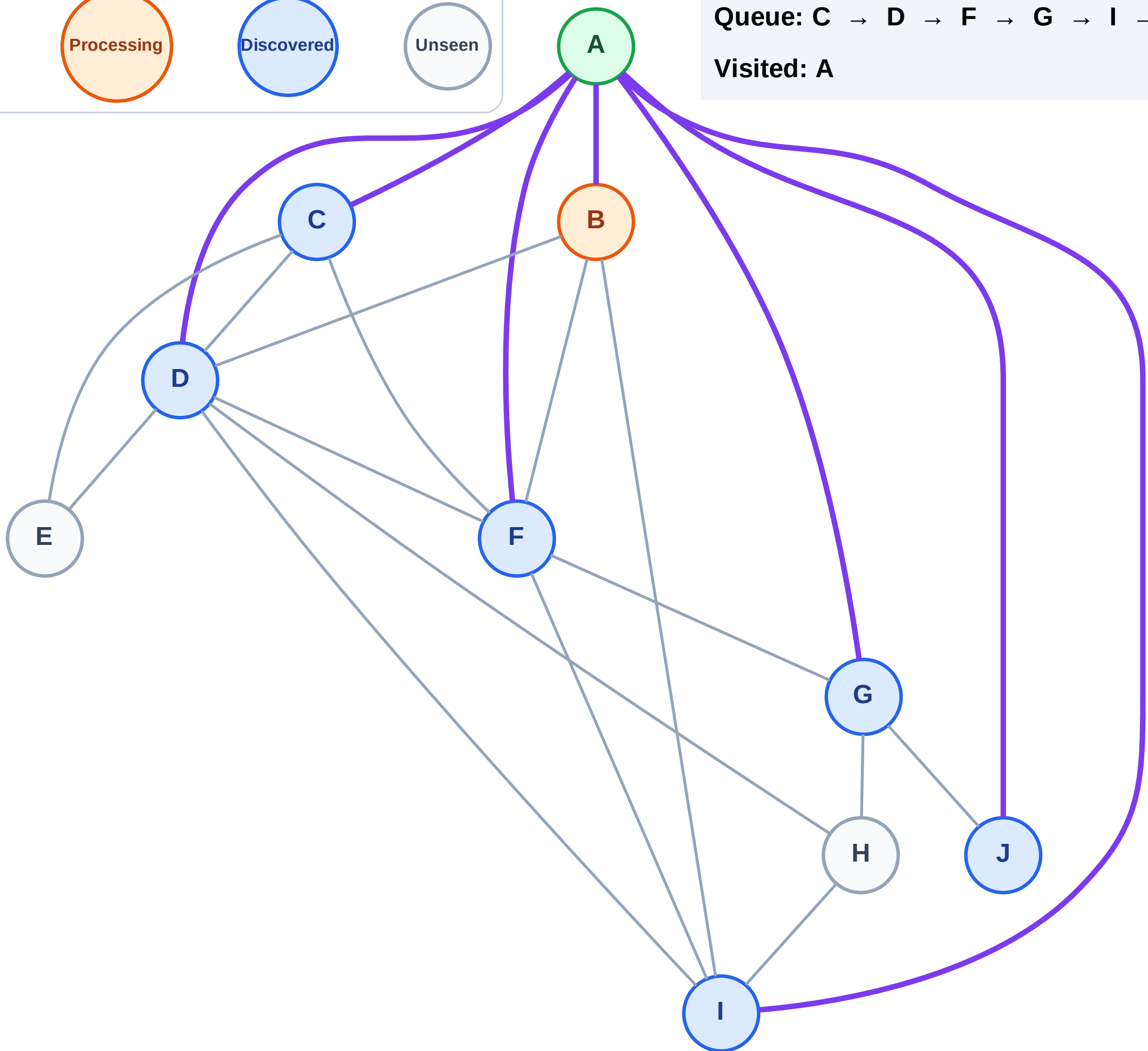
Discovered



Unseen

Queue: C → D → F → G → I → J

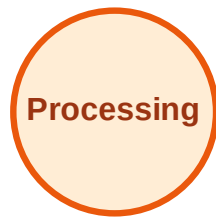
Visited: A



BFS Traversal

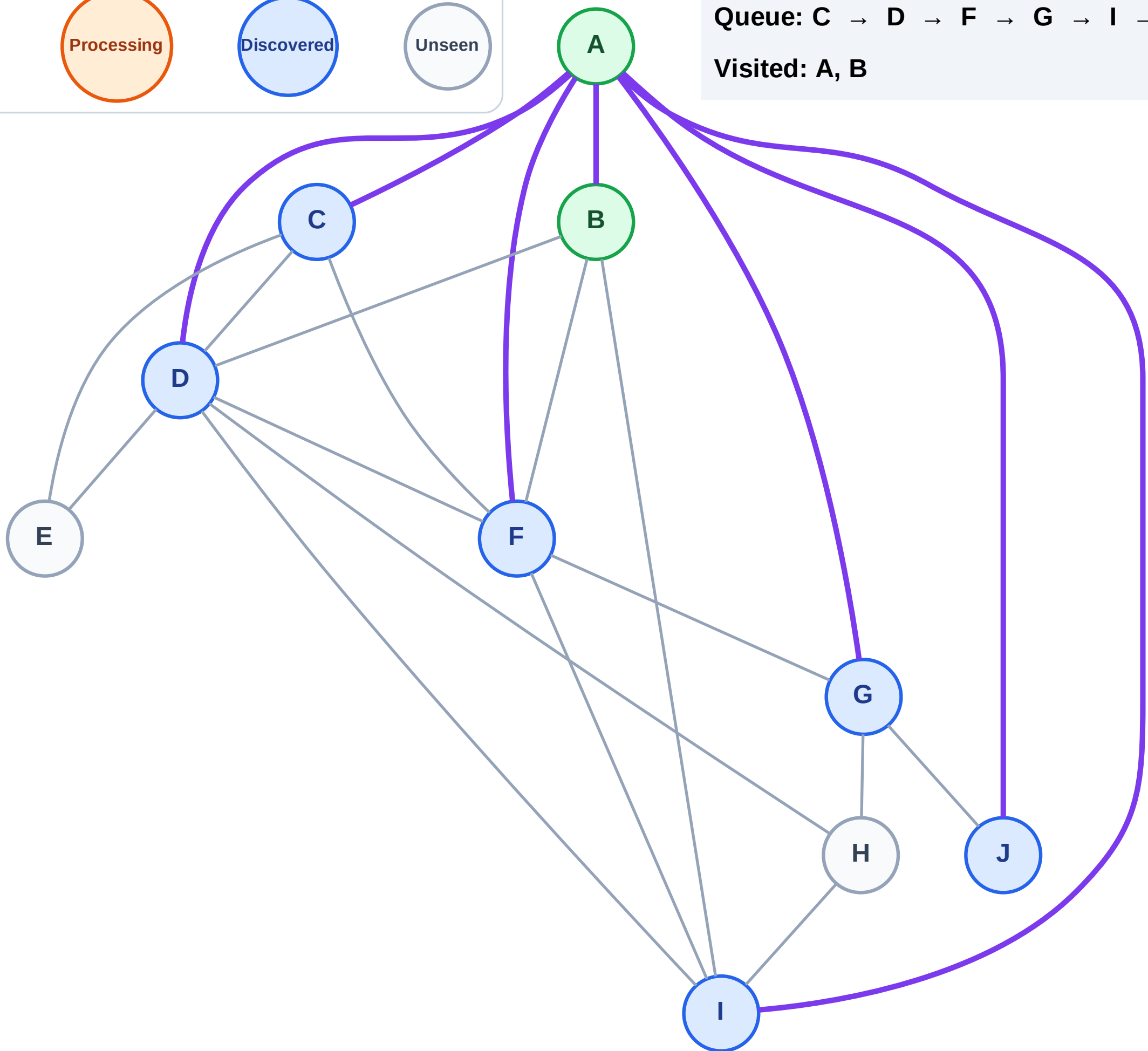
Step 5: No new neighbors from B

Legend



Queue: C → D → F → G → I → J

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

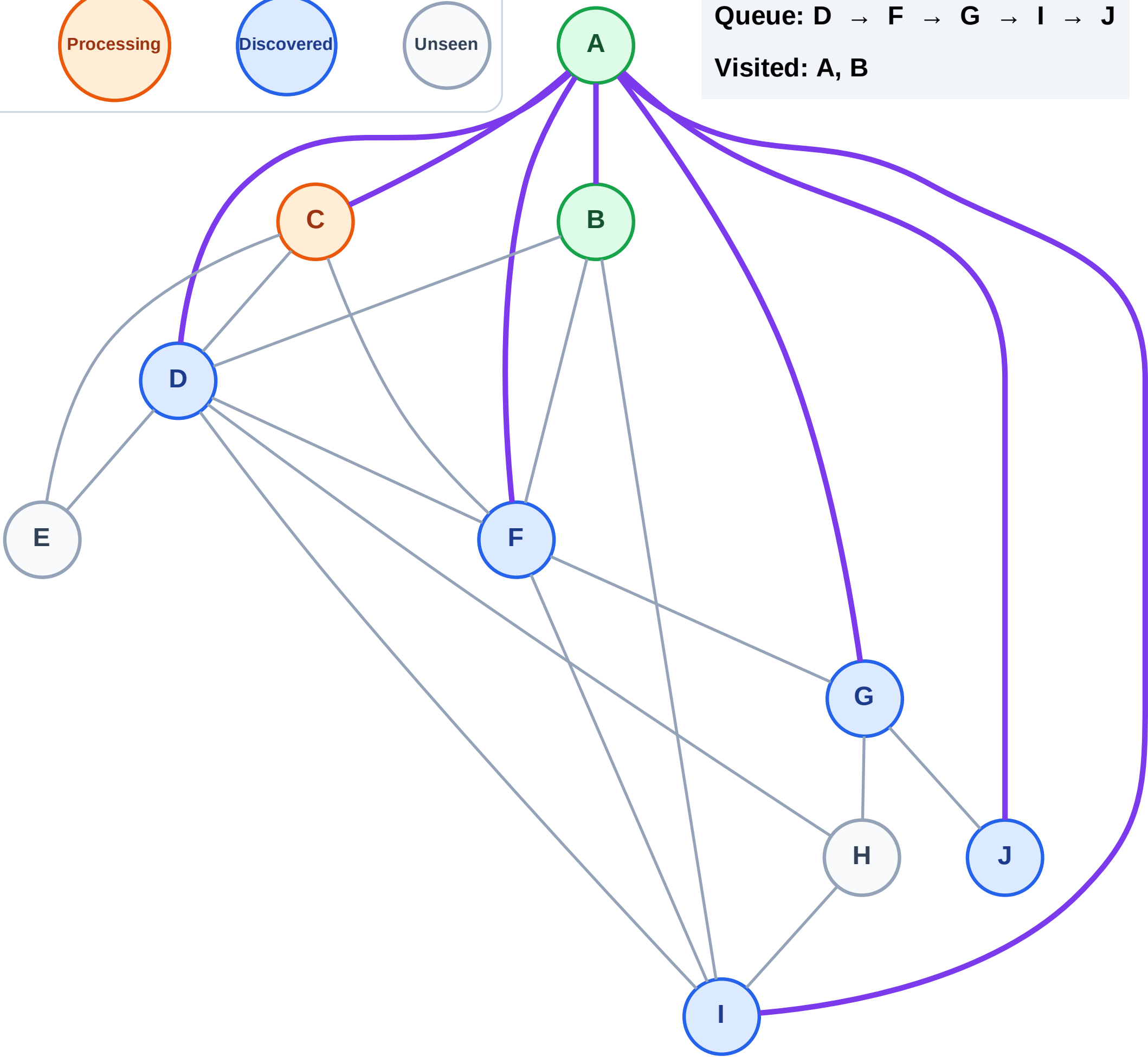
Processing

Discovered

Unseen

Queue: D → F → G → I → J

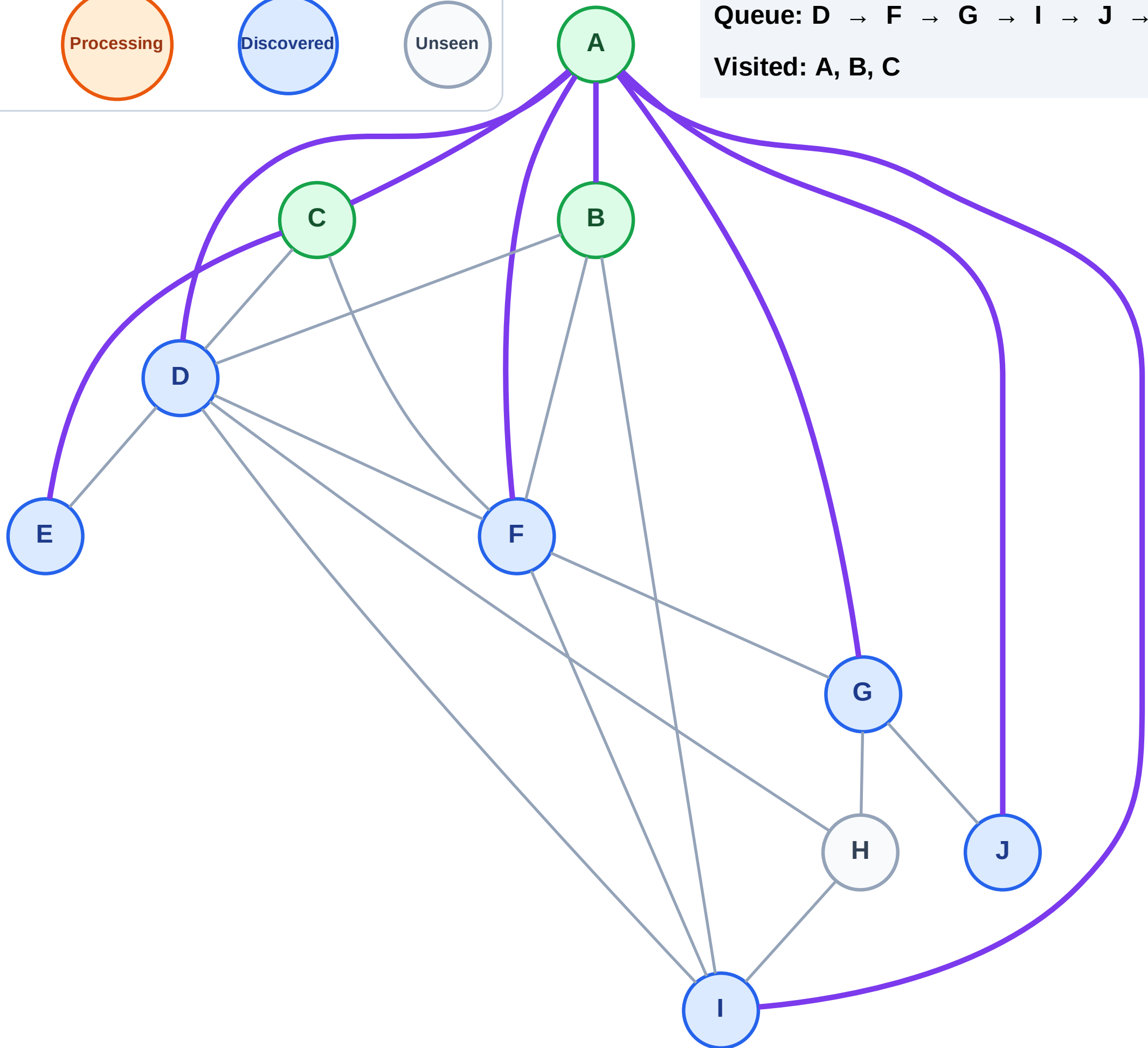
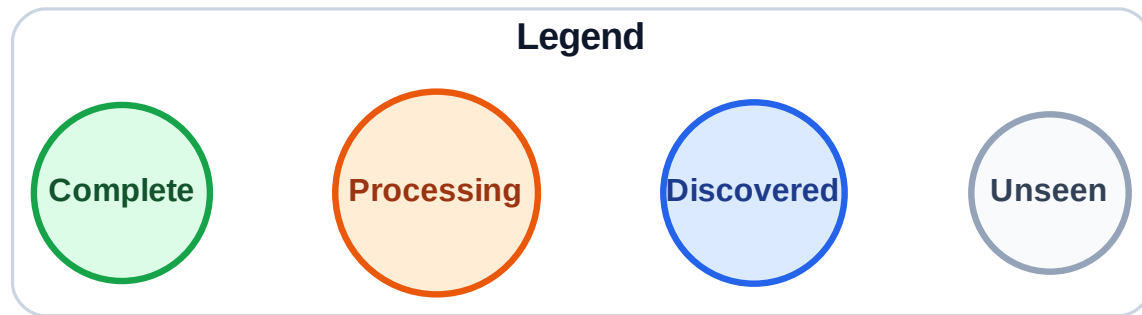
Visited: A, B



BFS Traversal

Step 7: Discover E from C

Queue: D → F → G → I → J → E
Visited: A, B, C



BFS Traversal

Step 8: Process node D

Legend



Complete



Processing



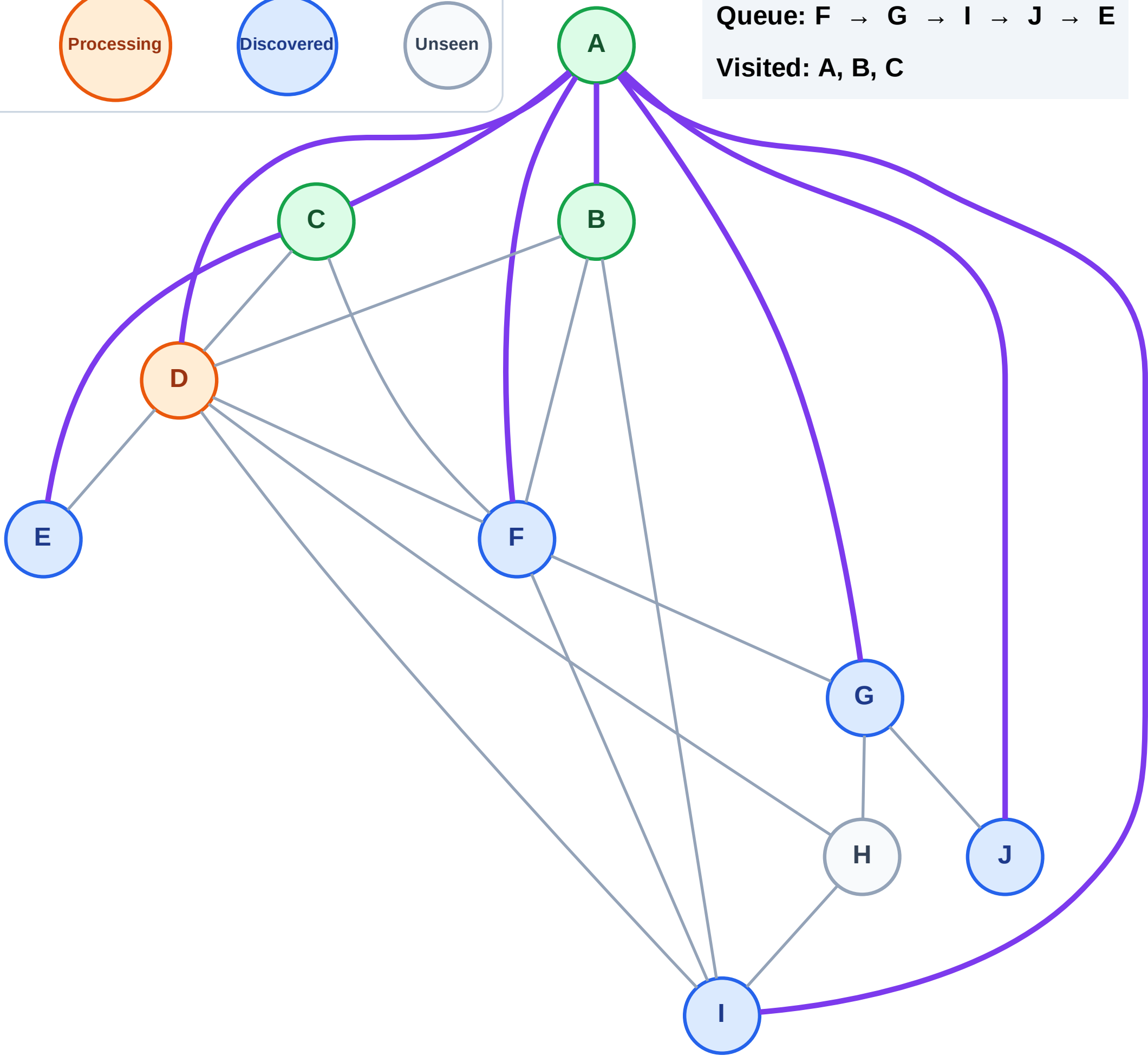
Discovered



Unseen

Queue: F → G → I → J → E

Visited: A, B, C

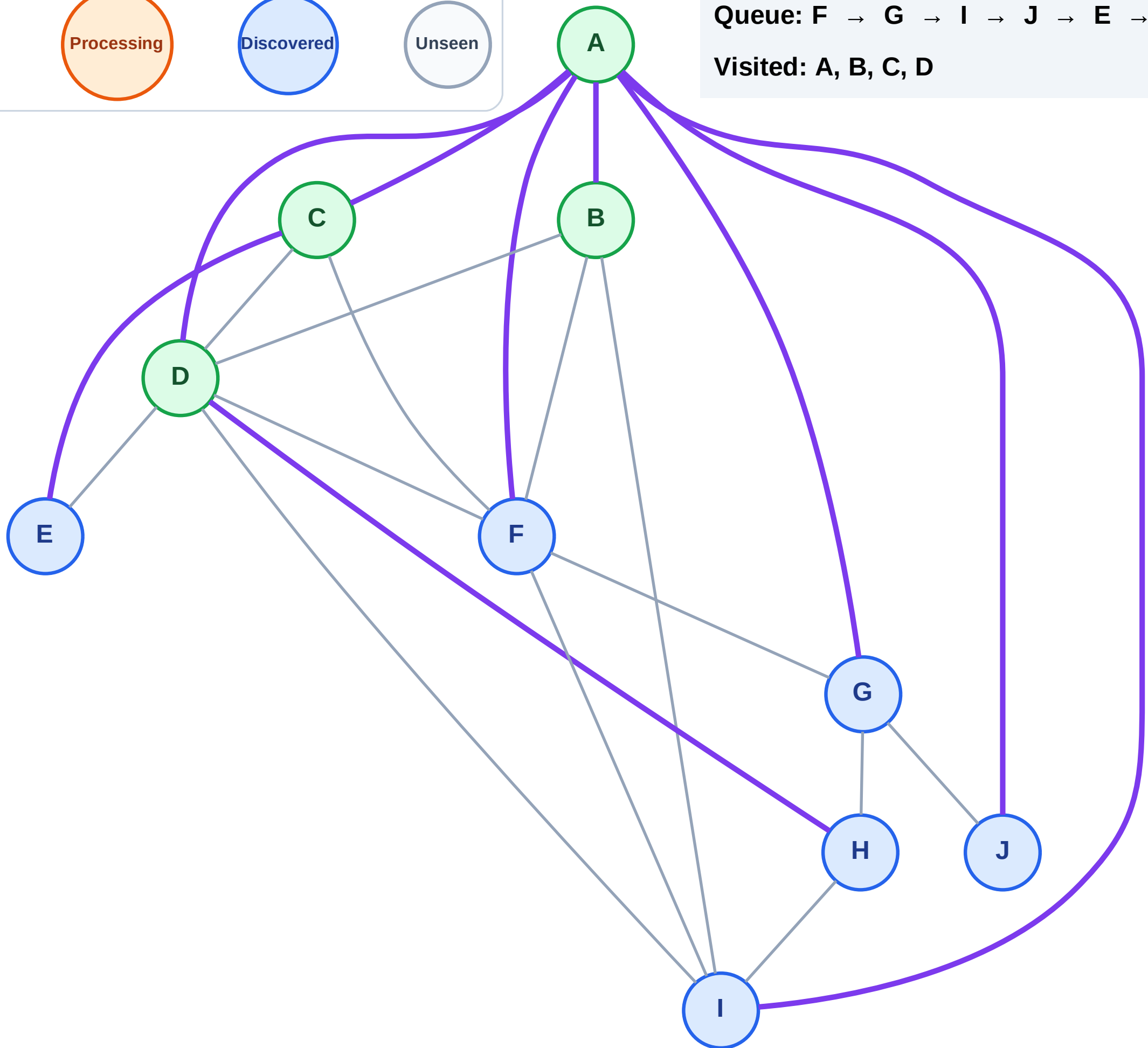
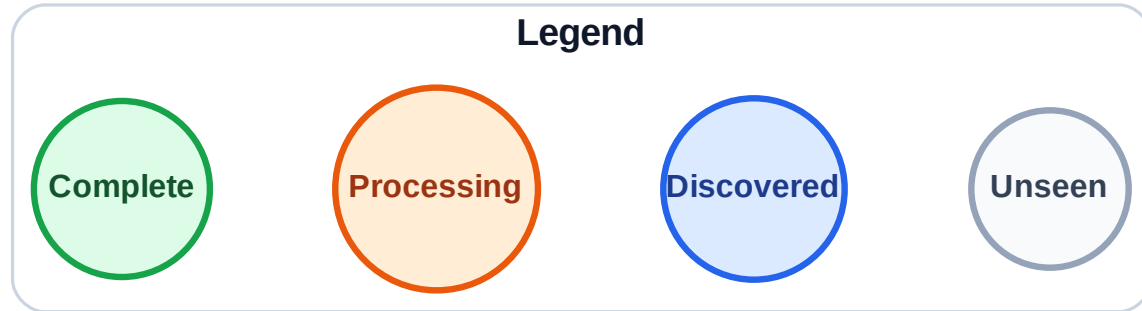


BFS Traversal

Step 9: Discover H from D

Queue: F → G → I → J → E → H

Visited: A, B, C, D



BFS Traversal

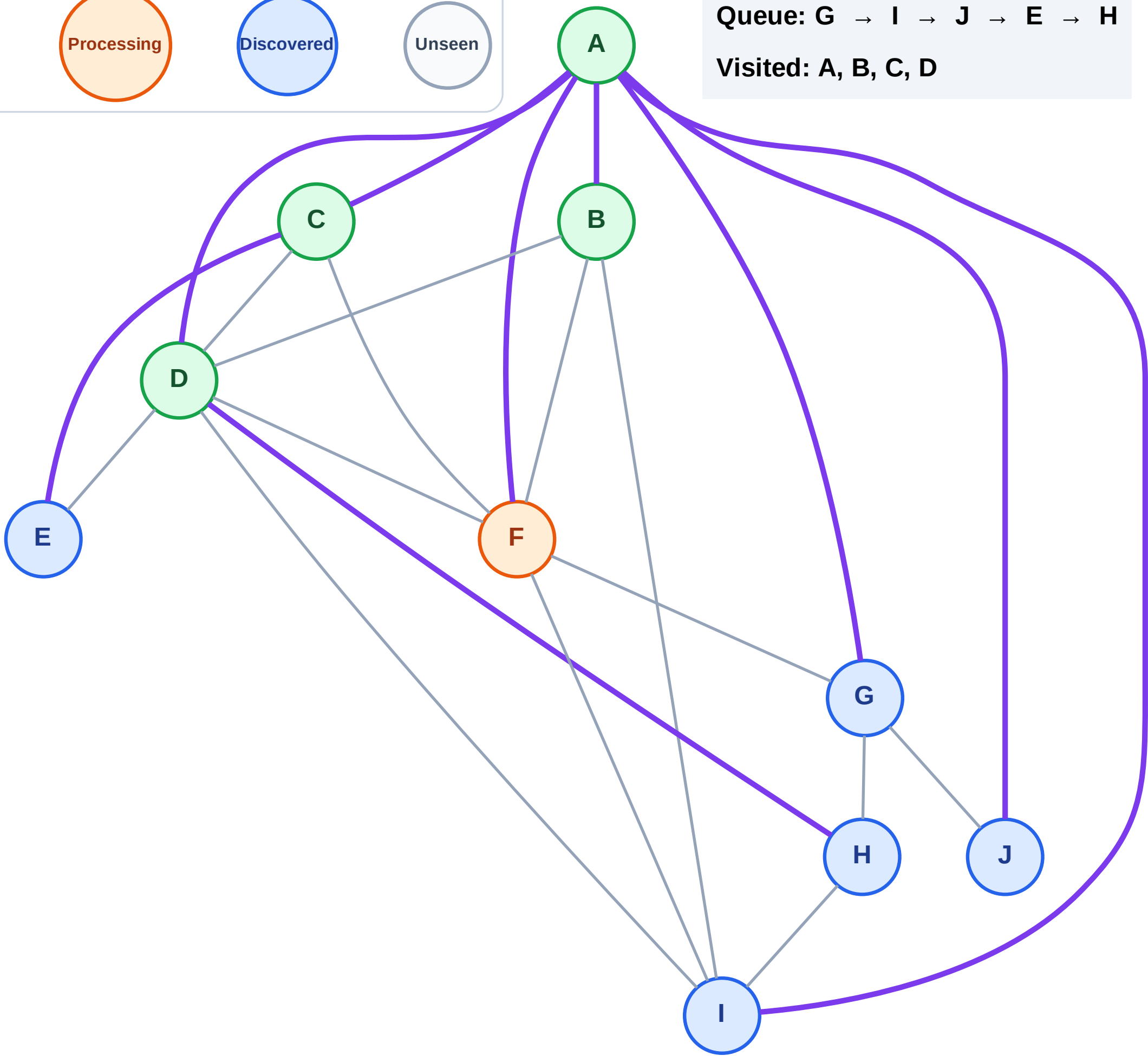
Step 10: Process node F

Legend



Queue: G → I → J → E → H

Visited: A, B, C, D



BFS Traversal

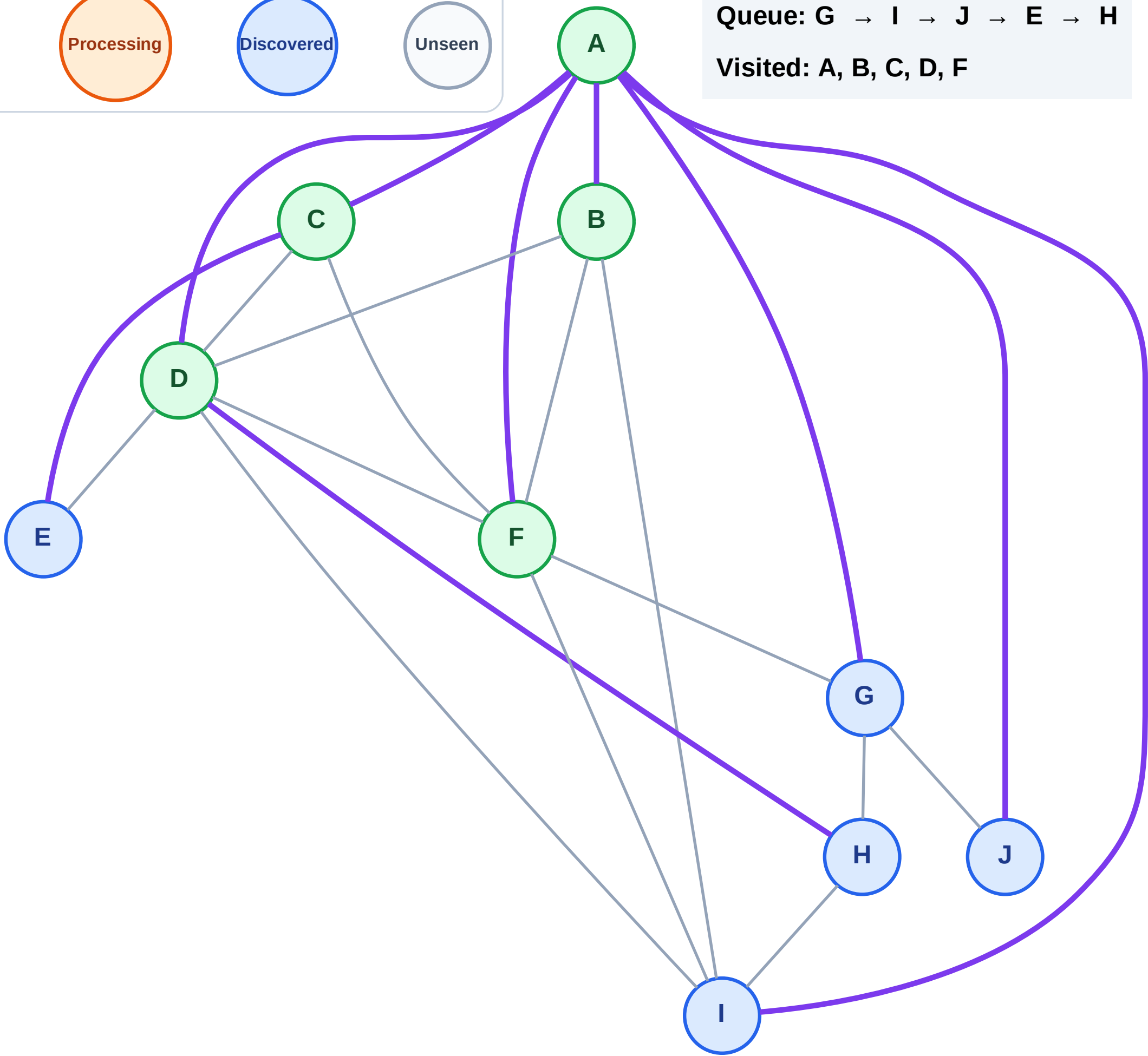
Step 11: No new neighbors from F

Legend



Queue: G → I → J → E → H

Visited: A, B, C, D, F



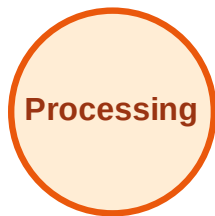
BFS Traversal

Step 12: Process node G

Legend



Complete



Processing



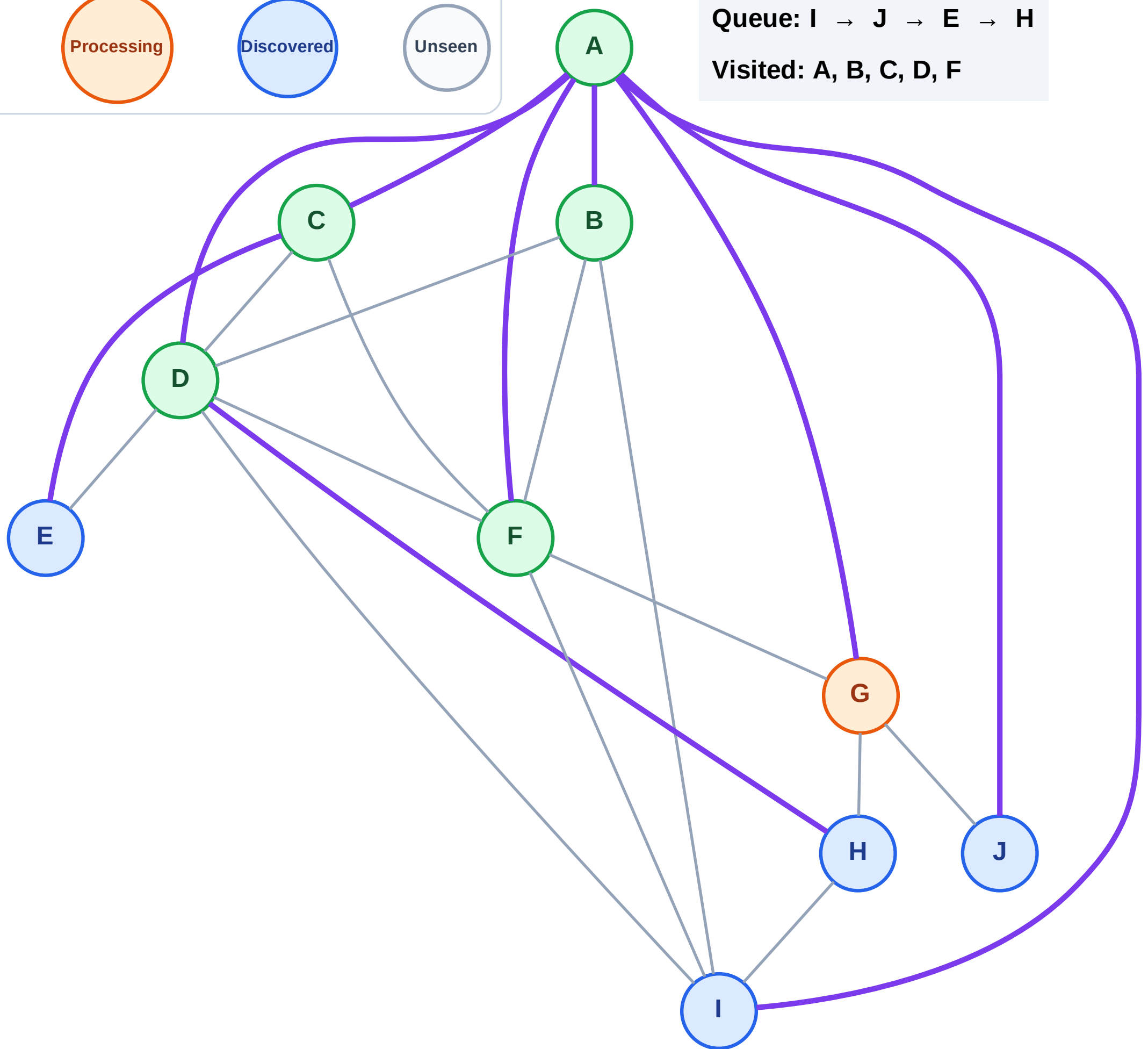
Discovered



Unseen

Queue: I → J → E → H

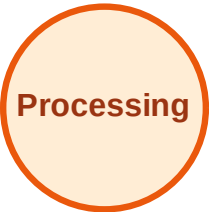
Visited: A, B, C, D, F



BFS Traversal

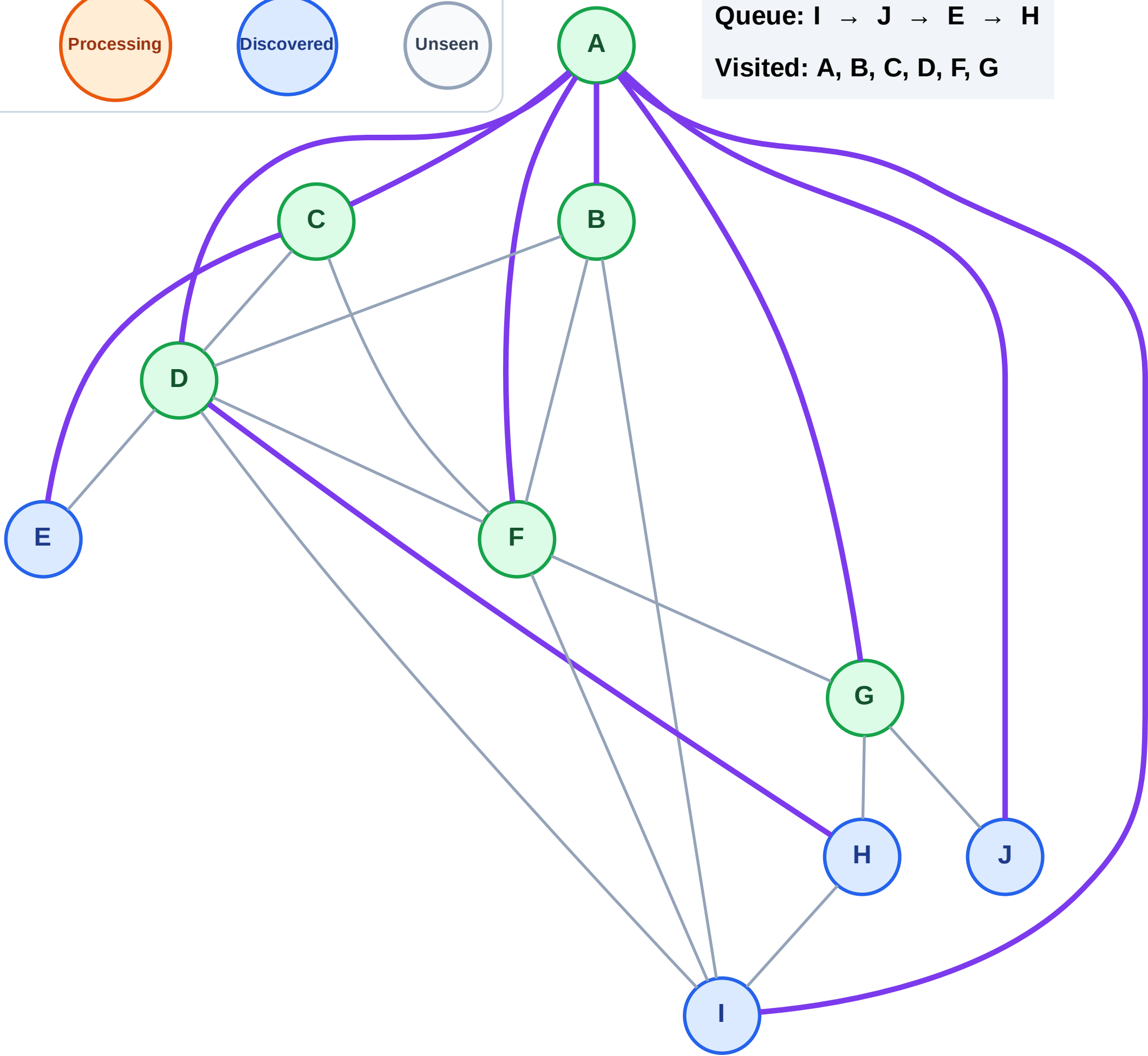
Step 13: No new neighbors from G

Legend



Queue: I → J → E → H

Visited: A, B, C, D, F, G



BFS Traversal

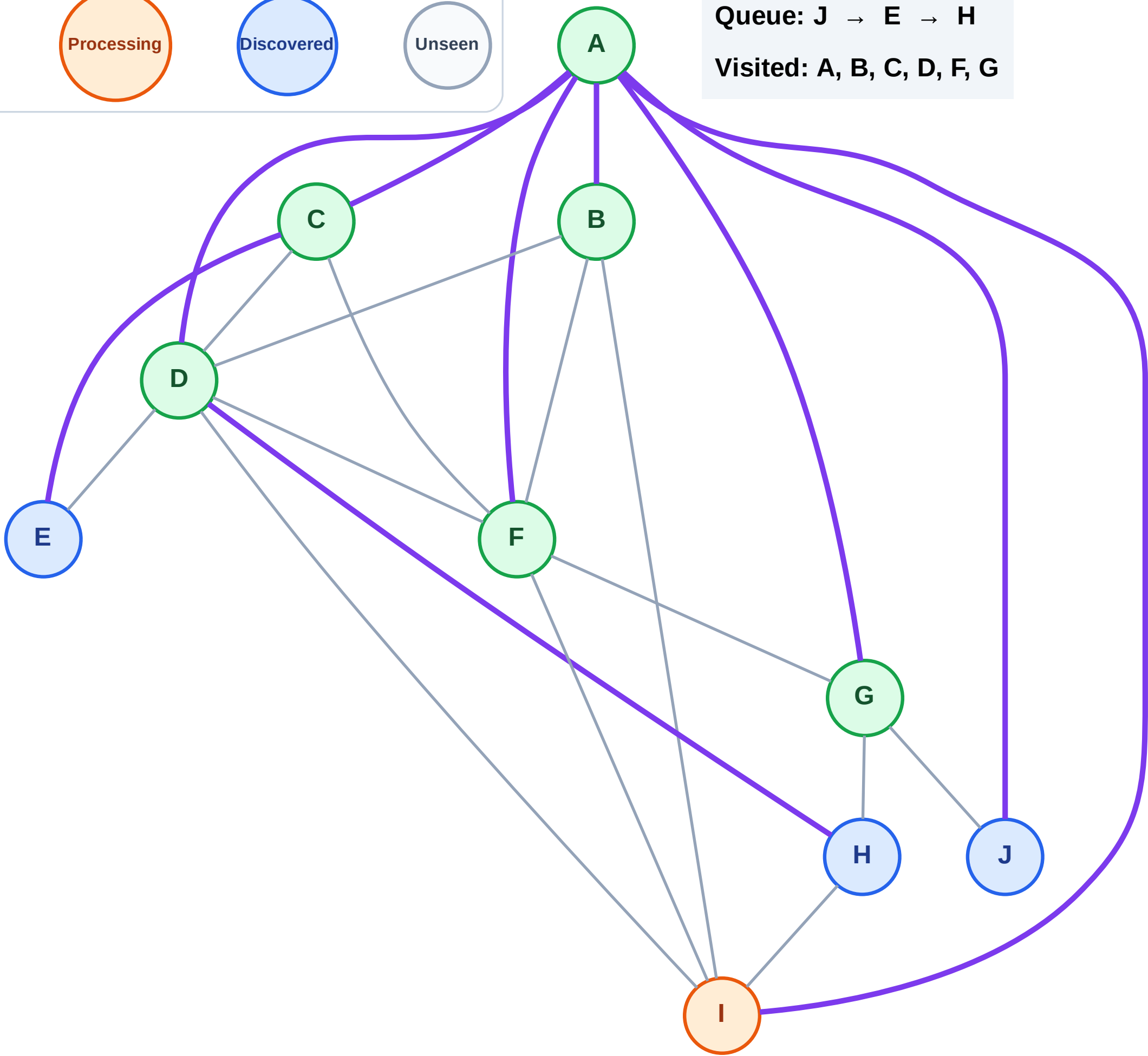
Step 14: Process node I

Legend



Queue: J → E → H

Visited: A, B, C, D, F, G



BFS Traversal

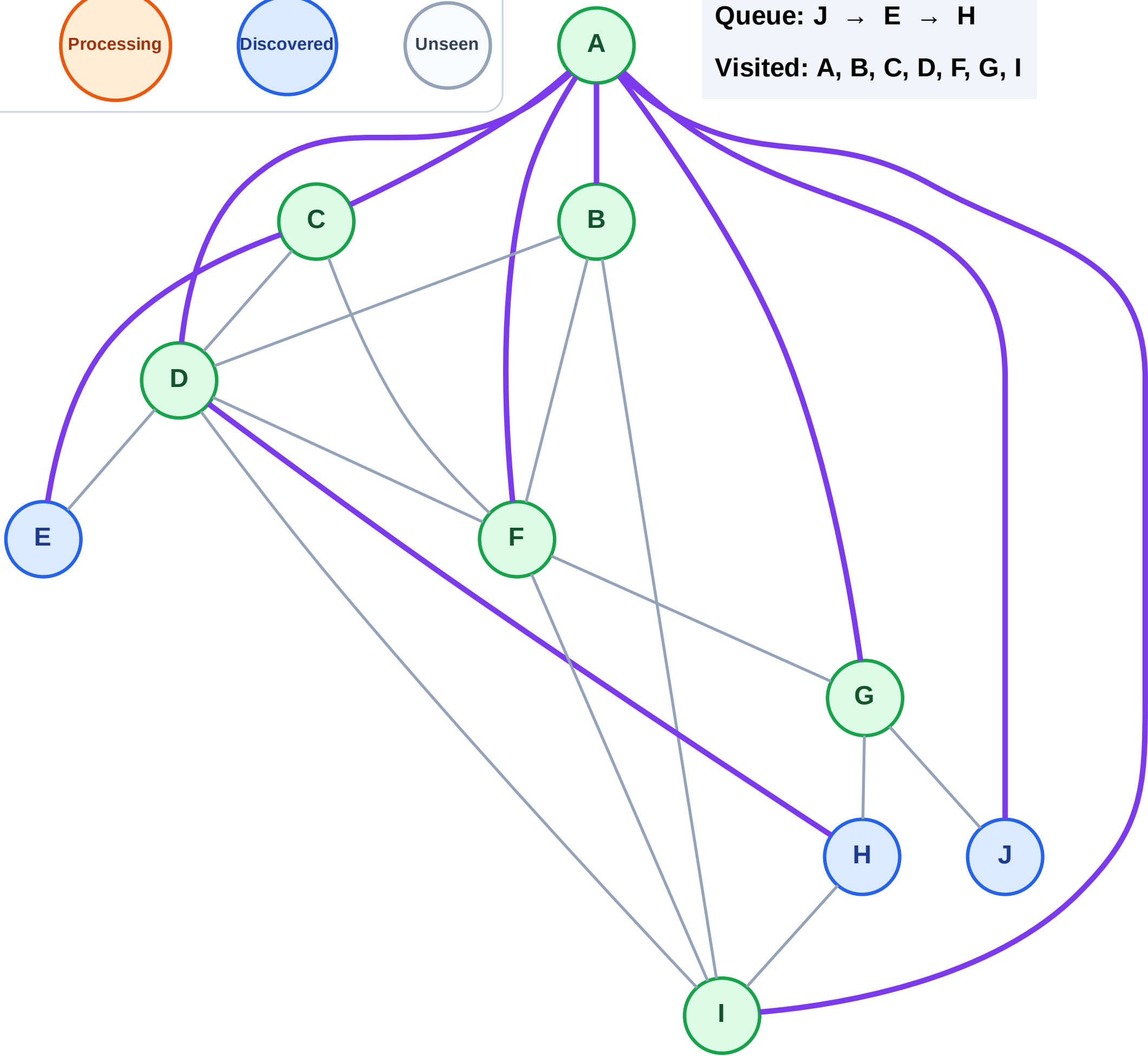
Step 15: No new neighbors from I

Legend



Queue: J → E → H

Visited: A, B, C, D, F, G, I



BFS Traversal

Step 16: Process node J

Legend

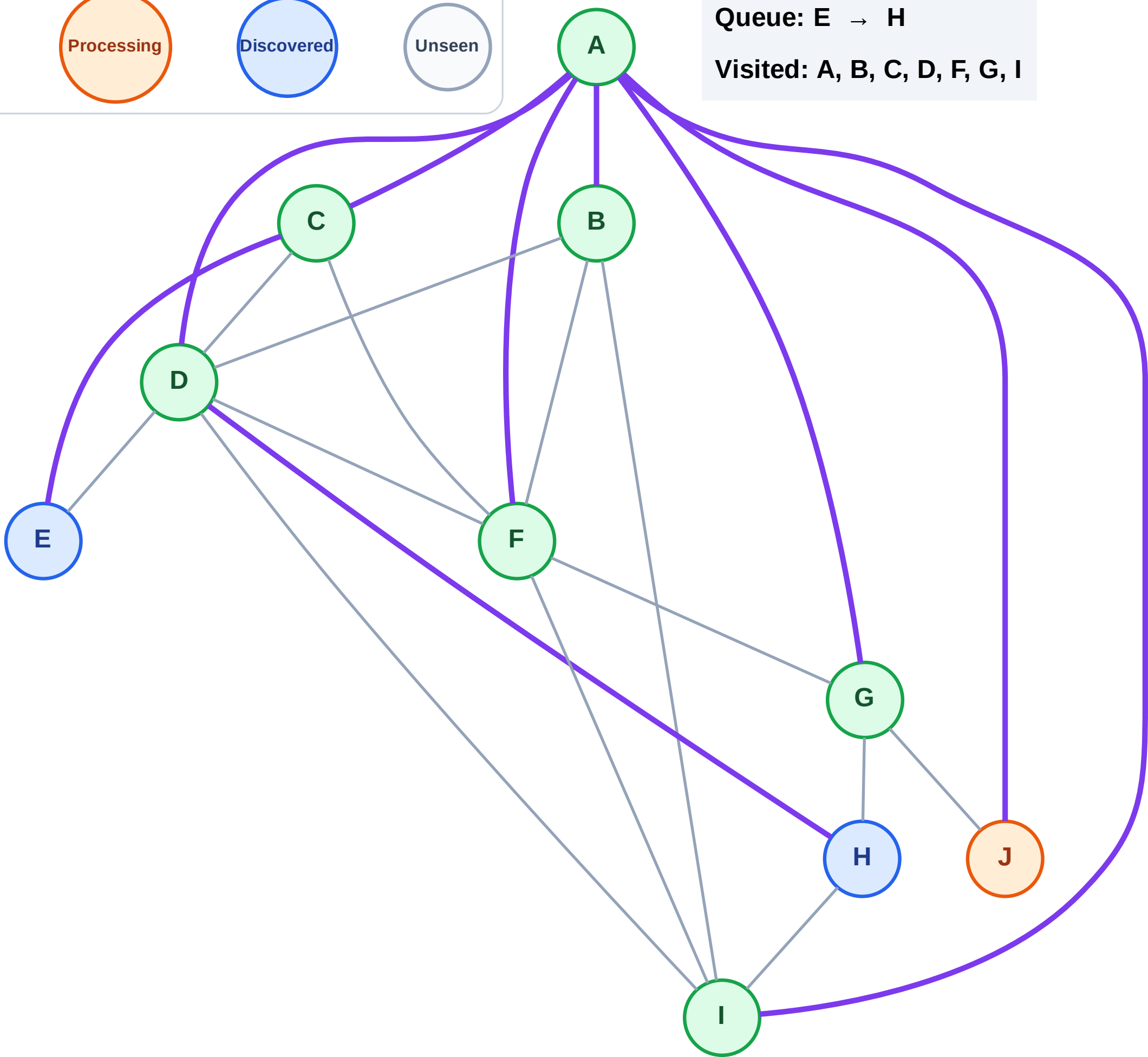
Complete

Processing

Discovered

Unseen

Queue: E → H
Visited: A, B, C, D, F, G, I



BFS Traversal

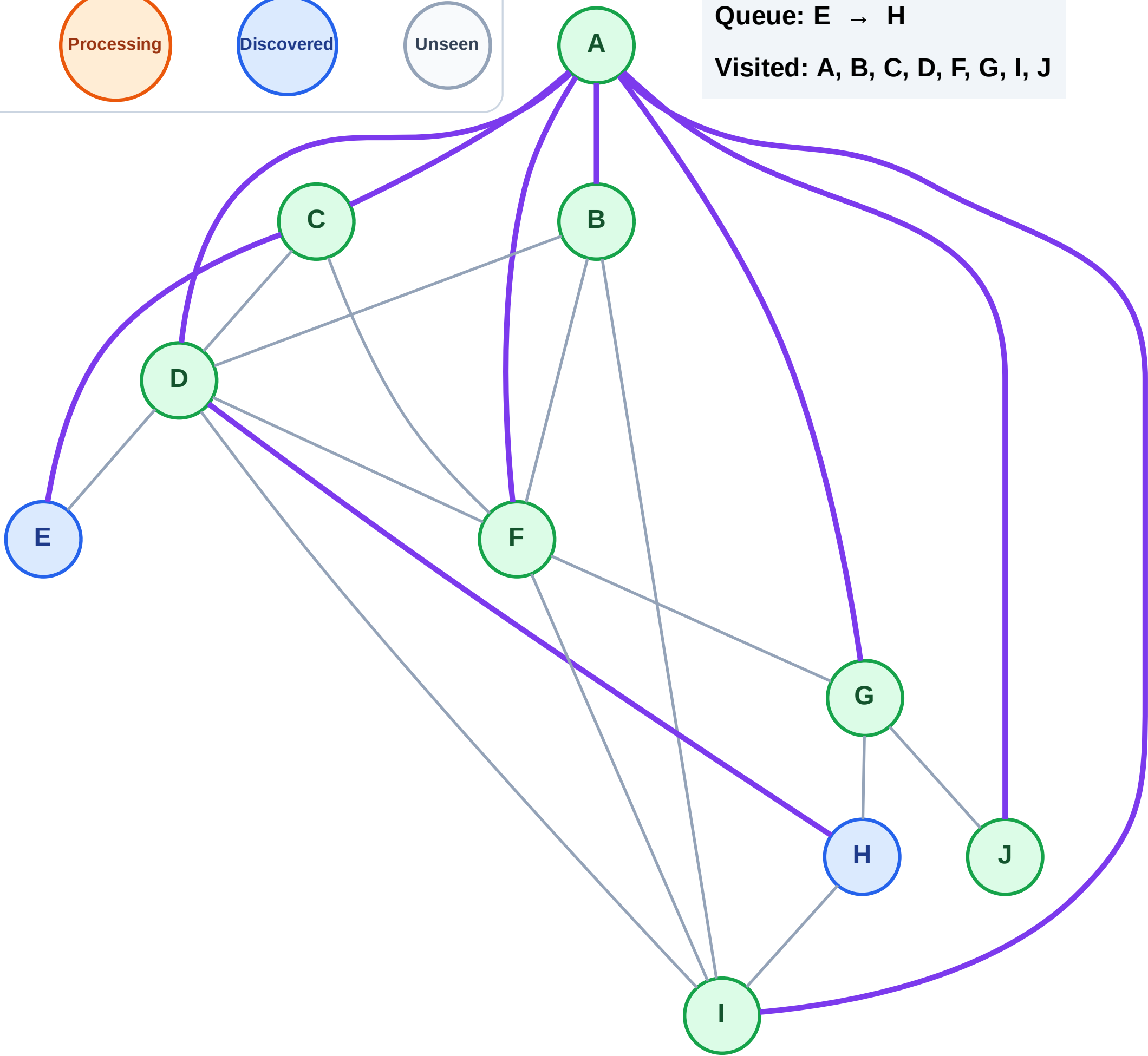
Step 17: No new neighbors from J

Legend



Queue: E → H

Visited: A, B, C, D, F, G, I, J



BFS Traversal

Step 18: Process node E

Legend



Complete



Processing



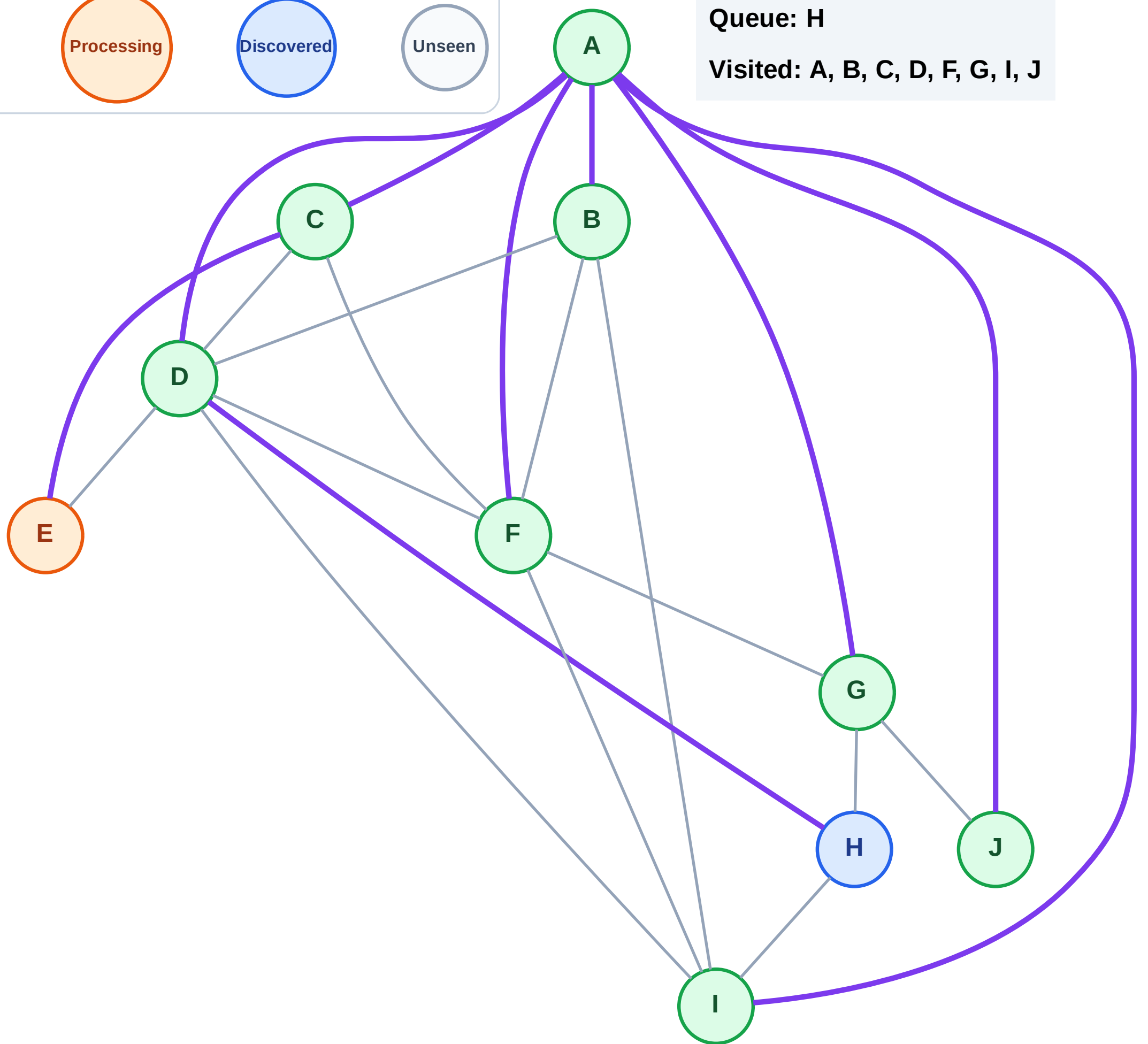
Discovered



Unseen

Queue: H

Visited: A, B, C, D, F, G, I, J



BFS Traversal

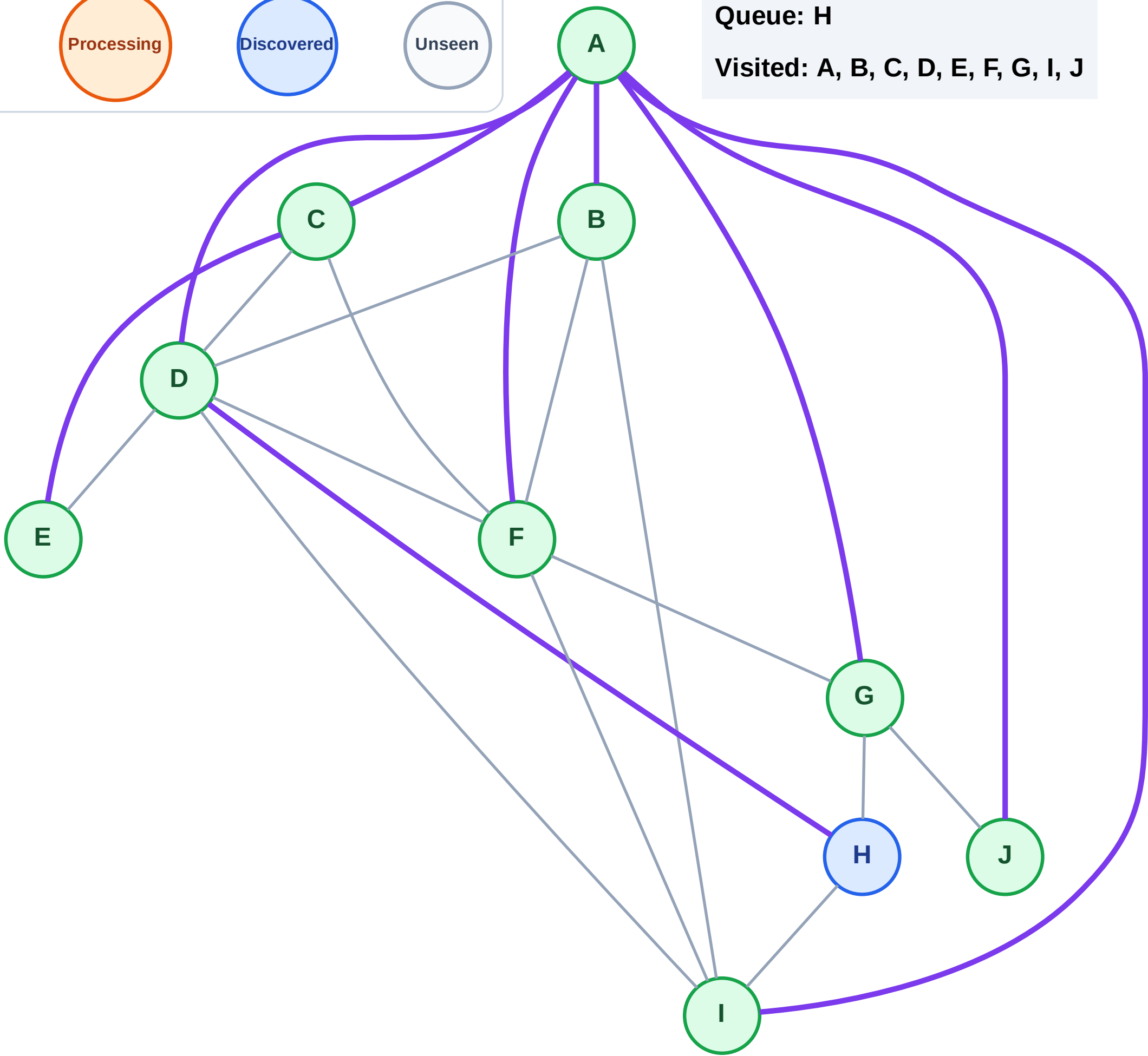
Step 19: No new neighbors from E

Legend



Queue: H

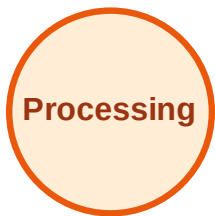
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

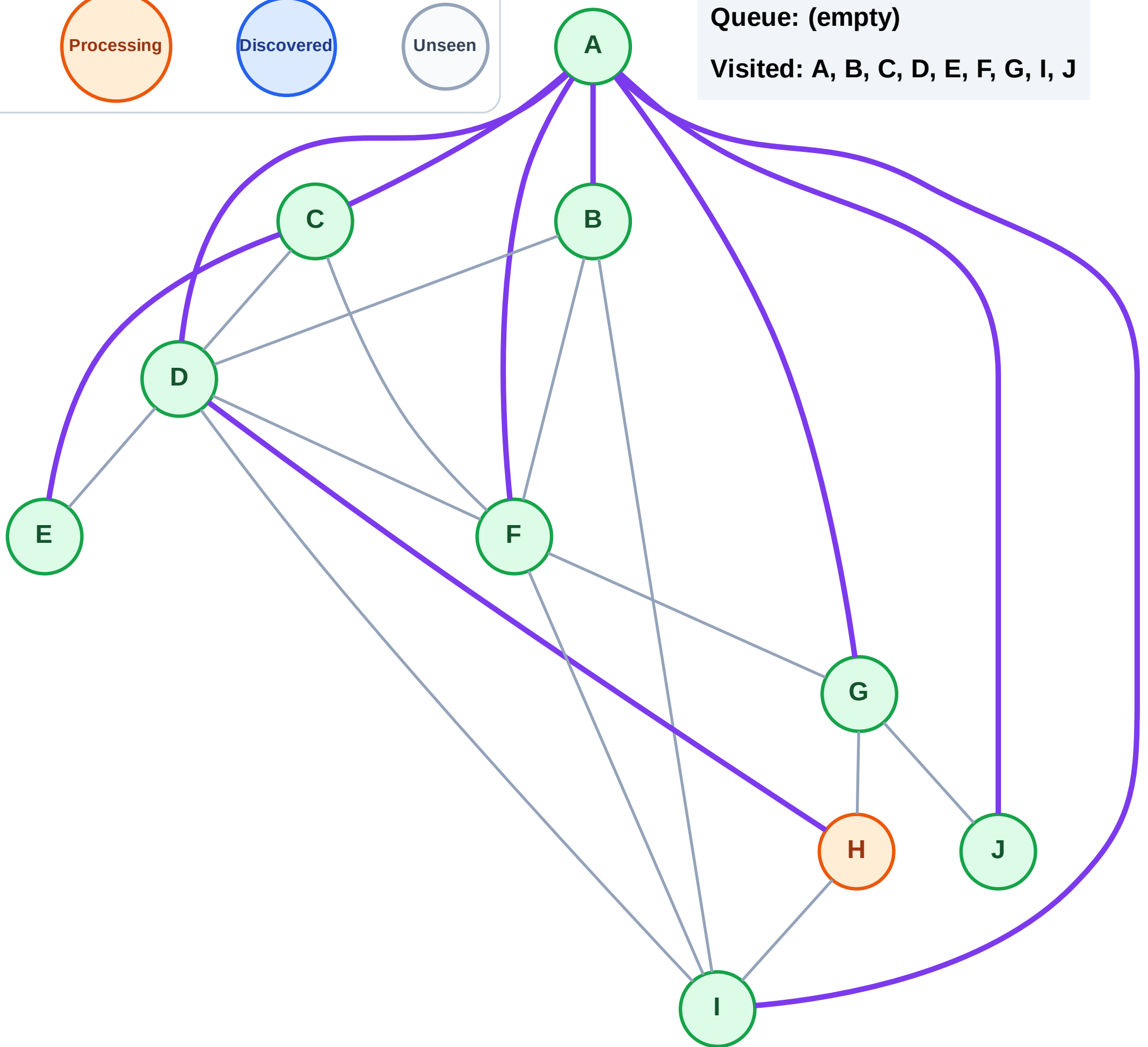
Step 20: Process node H

Legend



Queue: (empty)

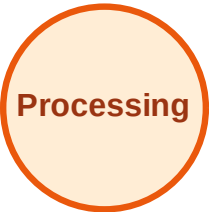
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

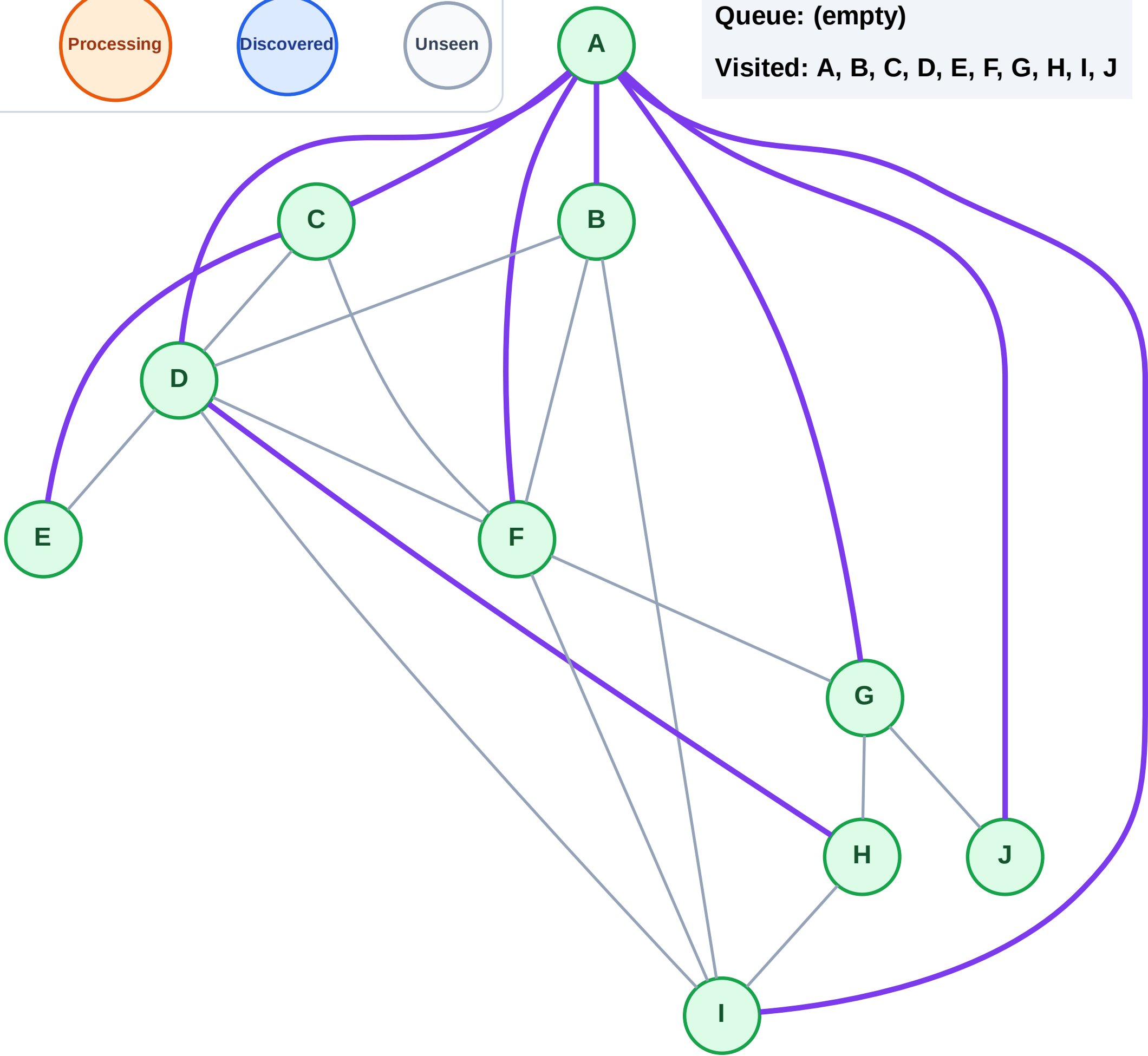
Step 21: No new neighbors from H

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

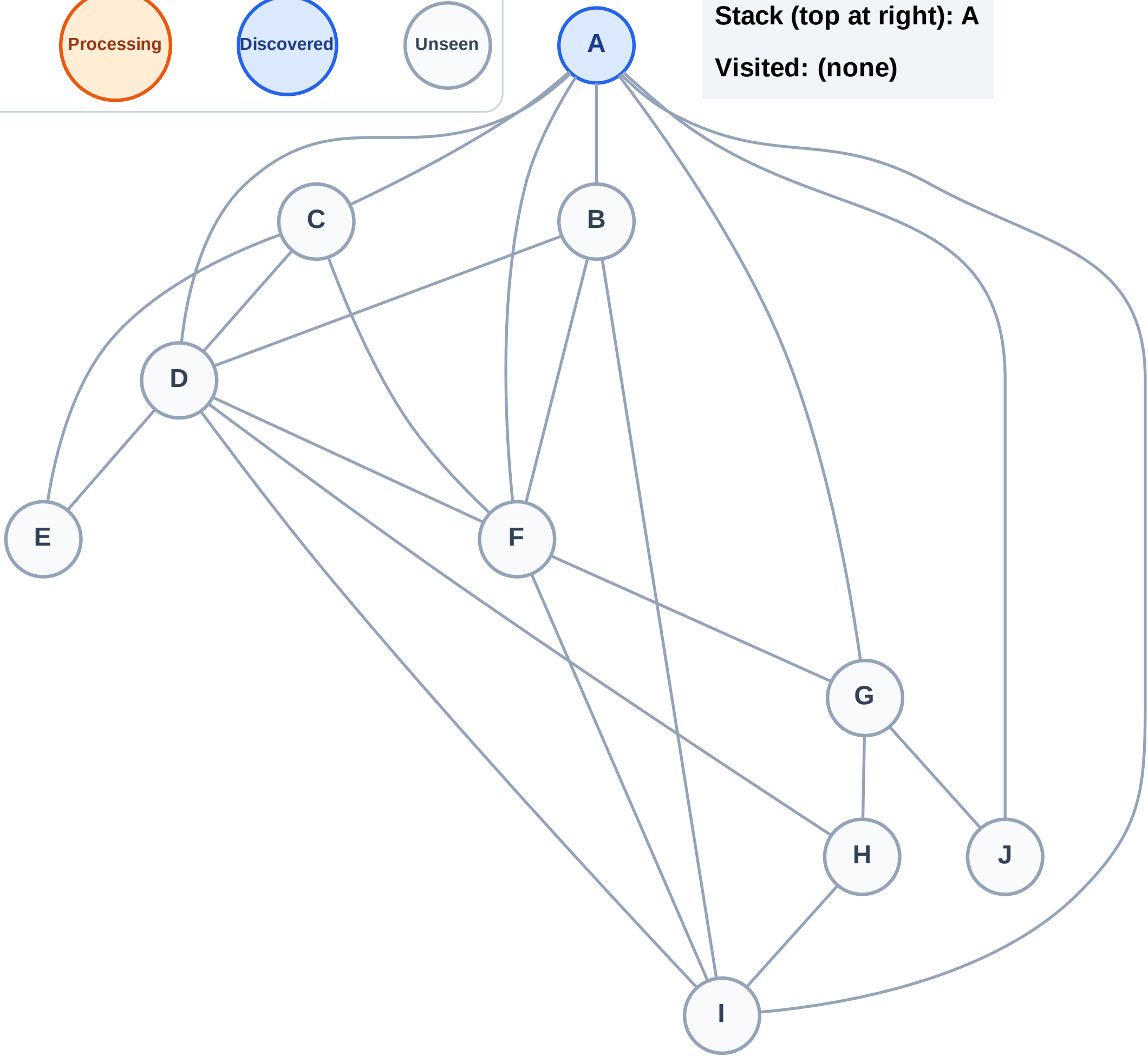
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

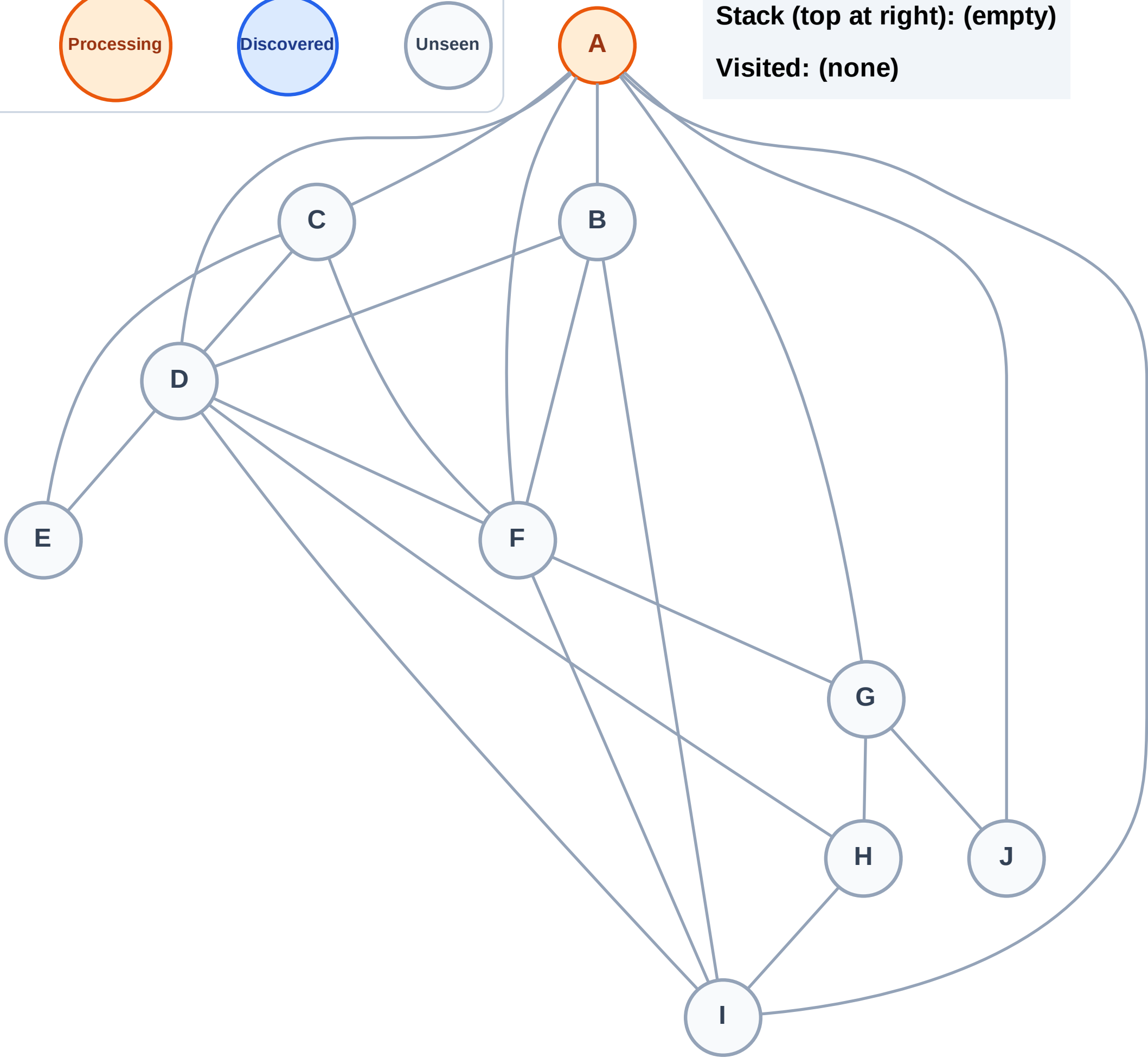
Step 2: Process node A

Legend



Stack (top at right): (empty)

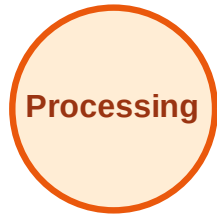
Visited: (none)



DFS Traversal

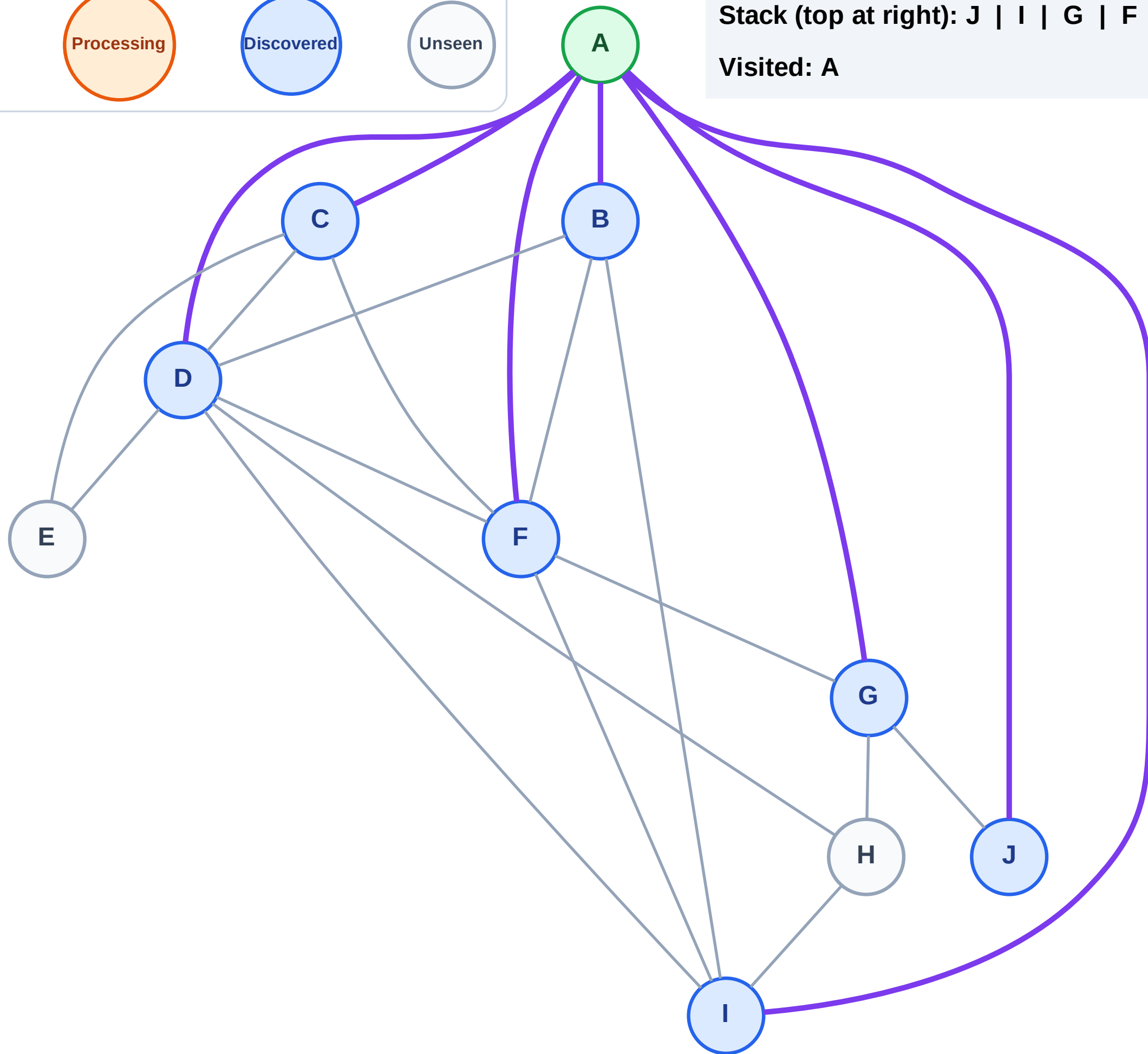
Step 3: Discover J, I, G, F, D, C, B from A

Legend



Stack (top at right): J | I | G | F | D | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

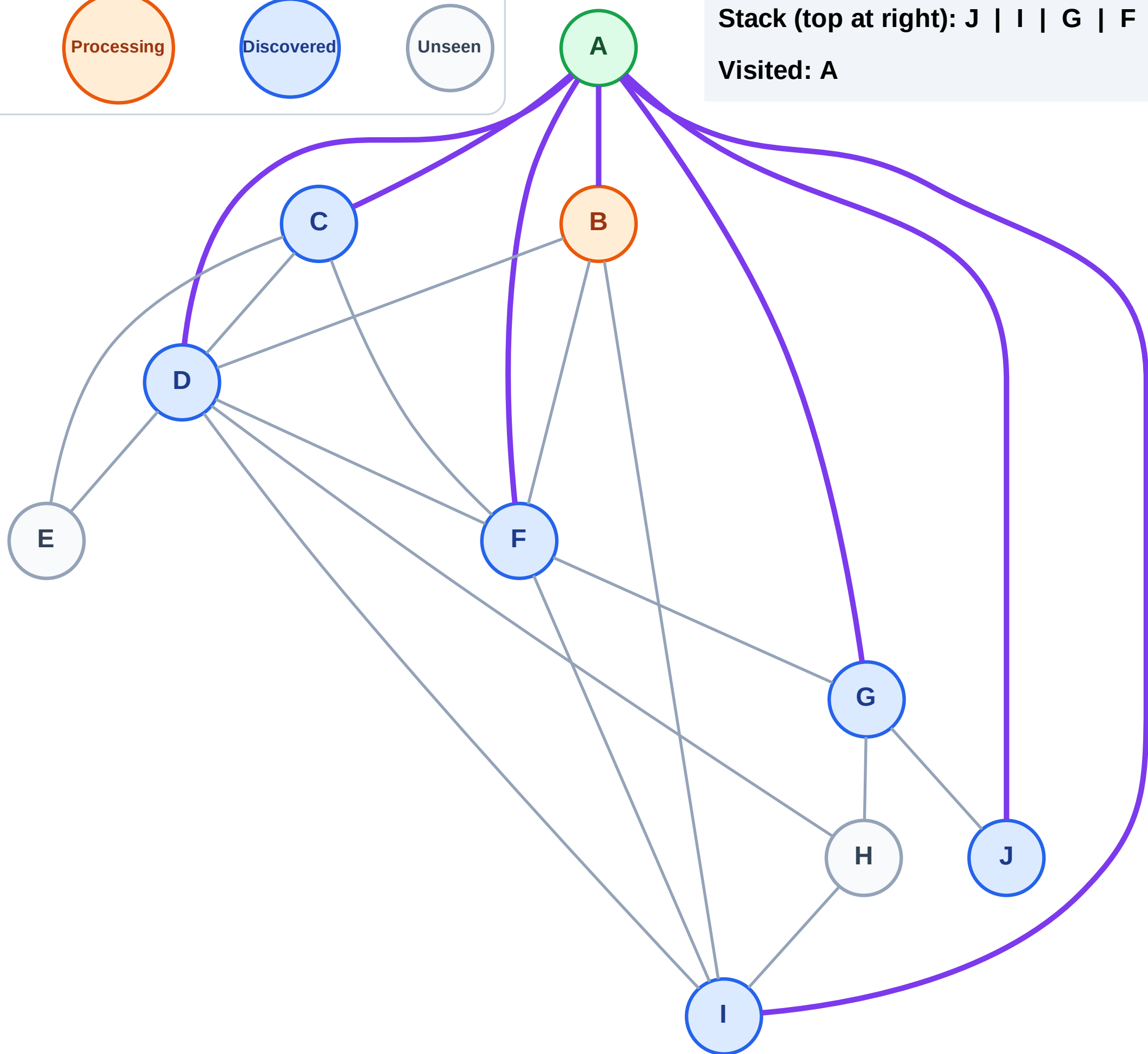
Processing

Discovered

Unseen

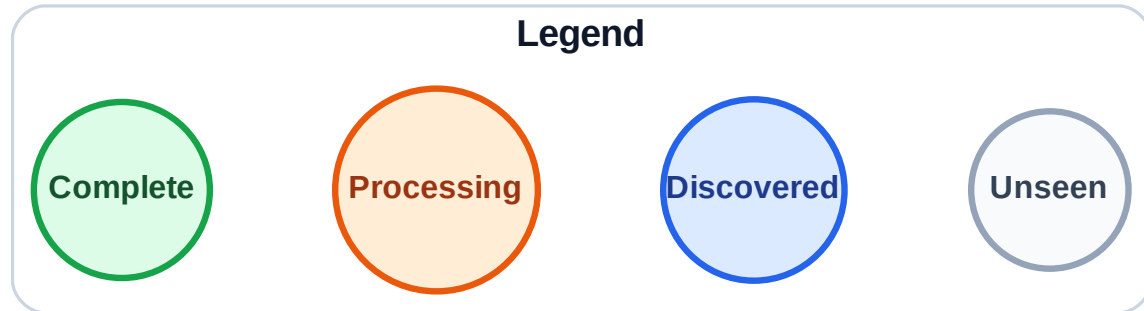
Stack (top at right): J | I | G | F | D | C

Visited: A

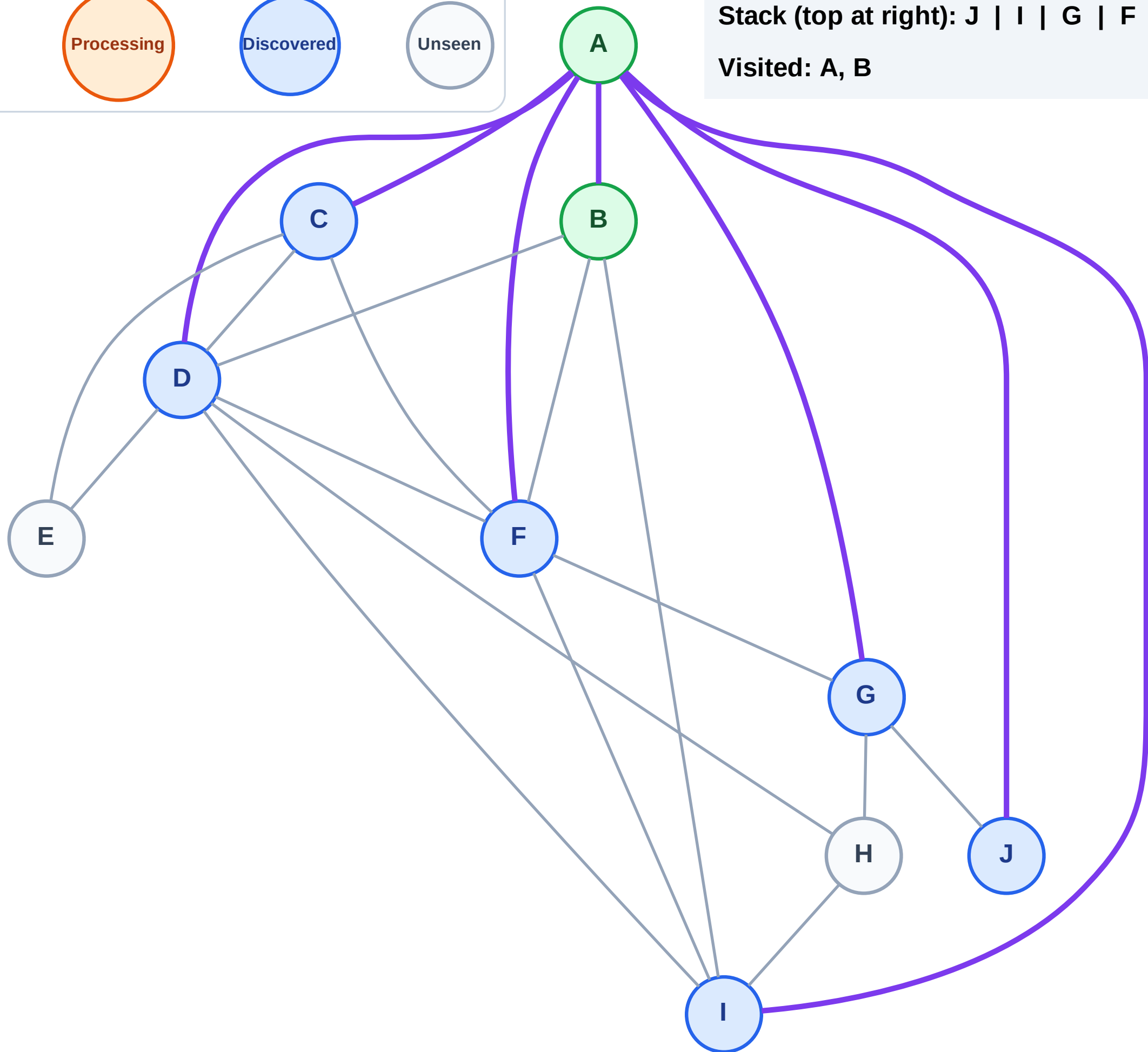


DFS Traversal

Step 5: No new neighbors from B



Stack (top at right): J | I | G | F | D | C
Visited: A, B



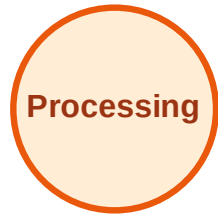
DFS Traversal

Step 6: Process node C

Legend



Complete



Processing



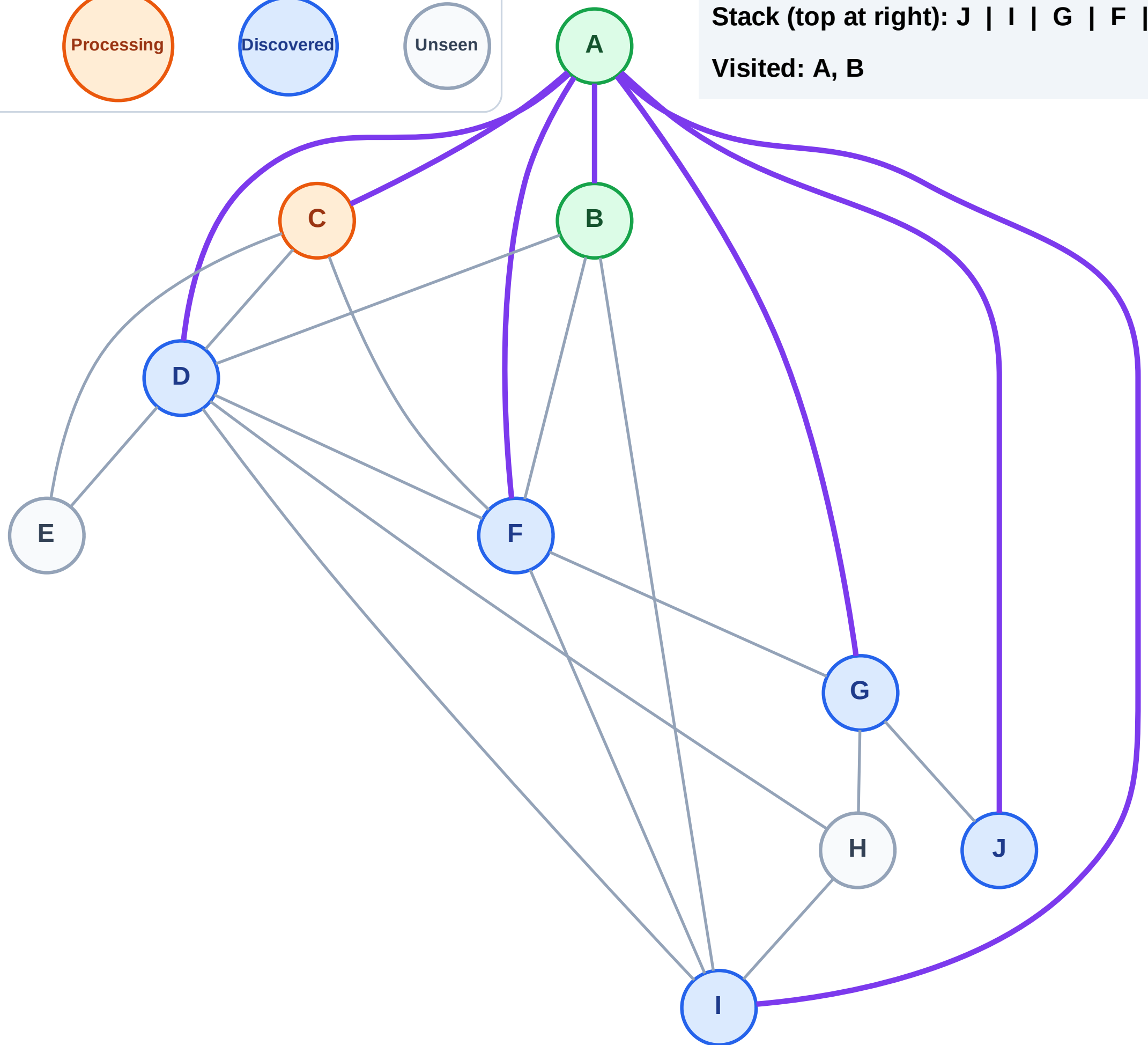
Discovered



Unseen

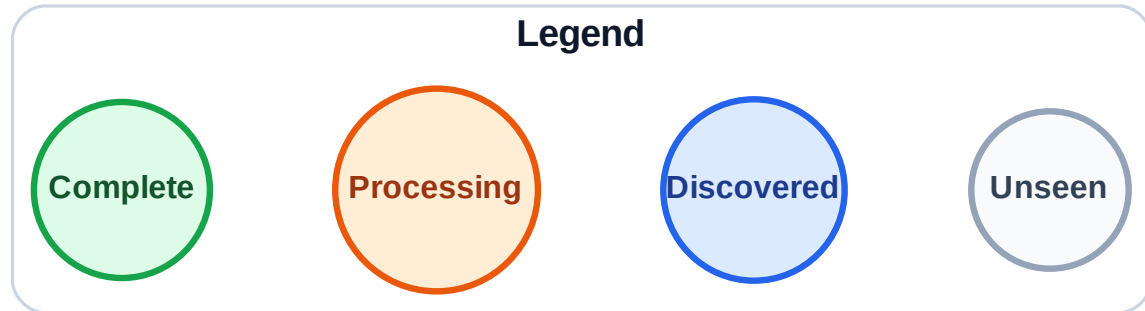
Stack (top at right): J | I | G | F | D

Visited: A, B

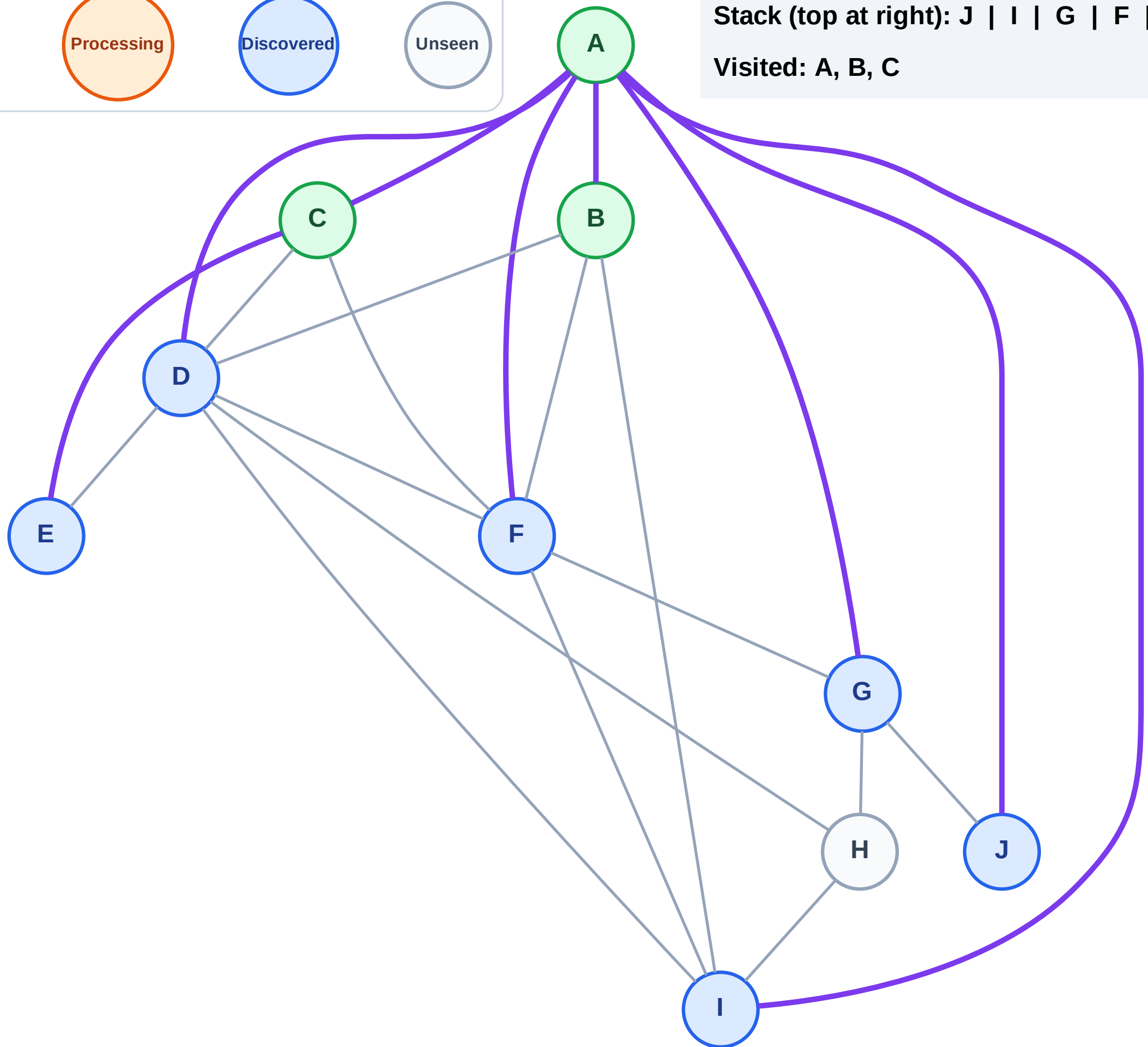


DFS Traversal

Step 7: Discover E from C



Stack (top at right): J | I | G | F | D | E
Visited: A, B, C

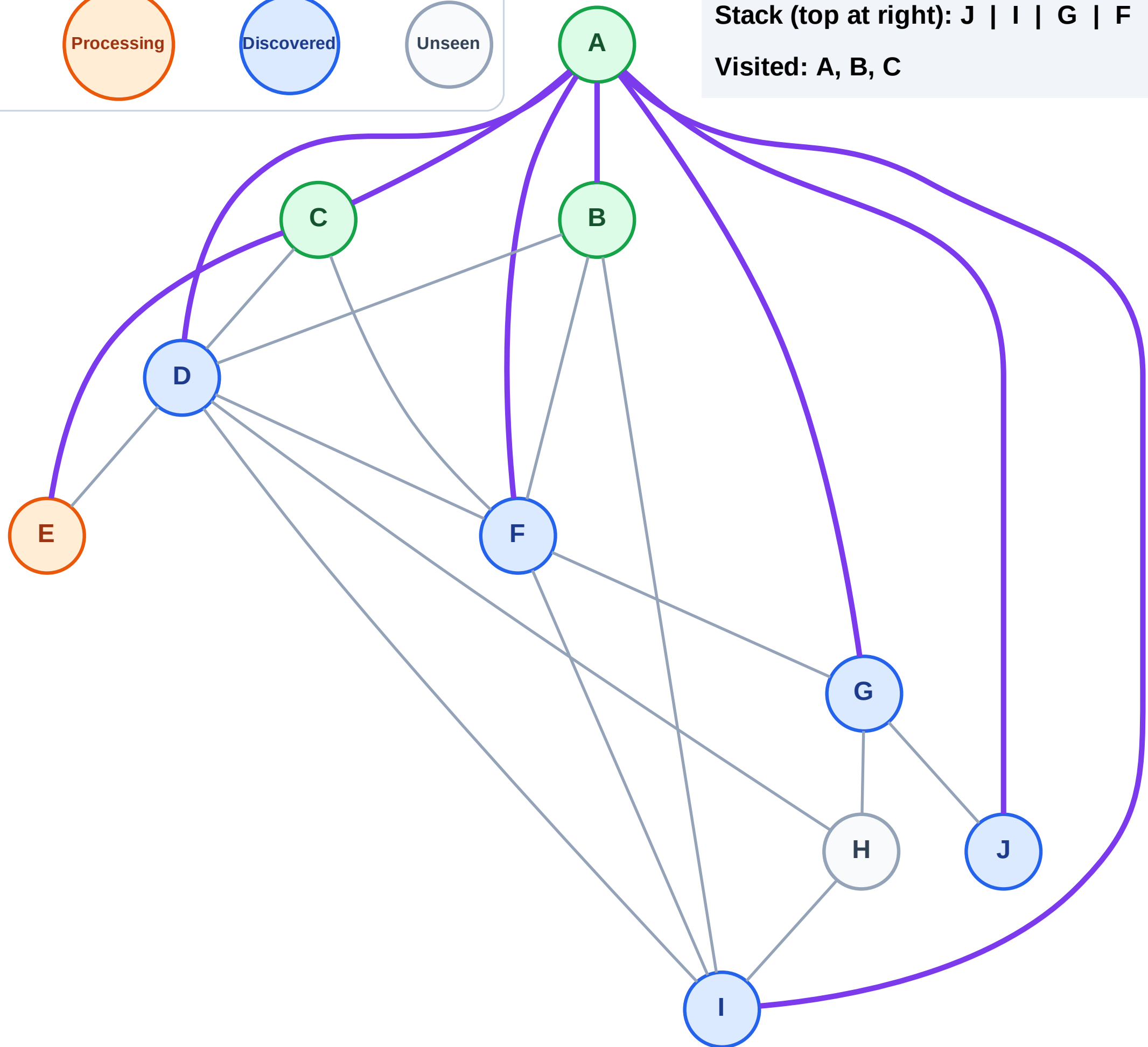
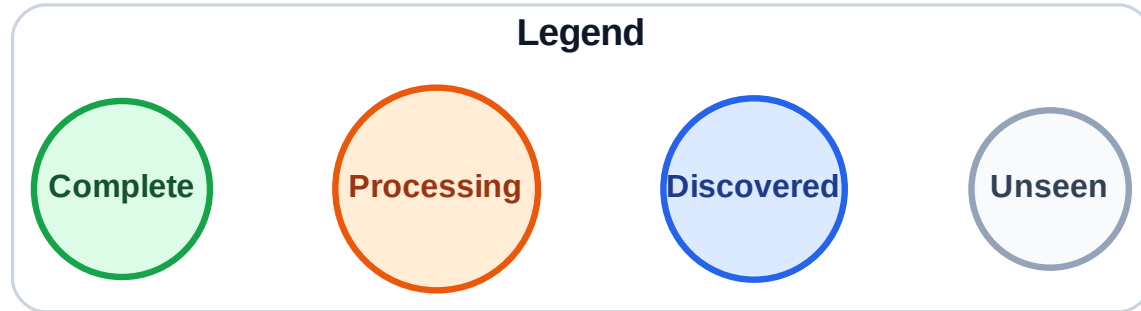


DFS Traversal

Step 8: Process node E

Stack (top at right): J | I | G | F | D

Visited: A, B, C



DFS Traversal

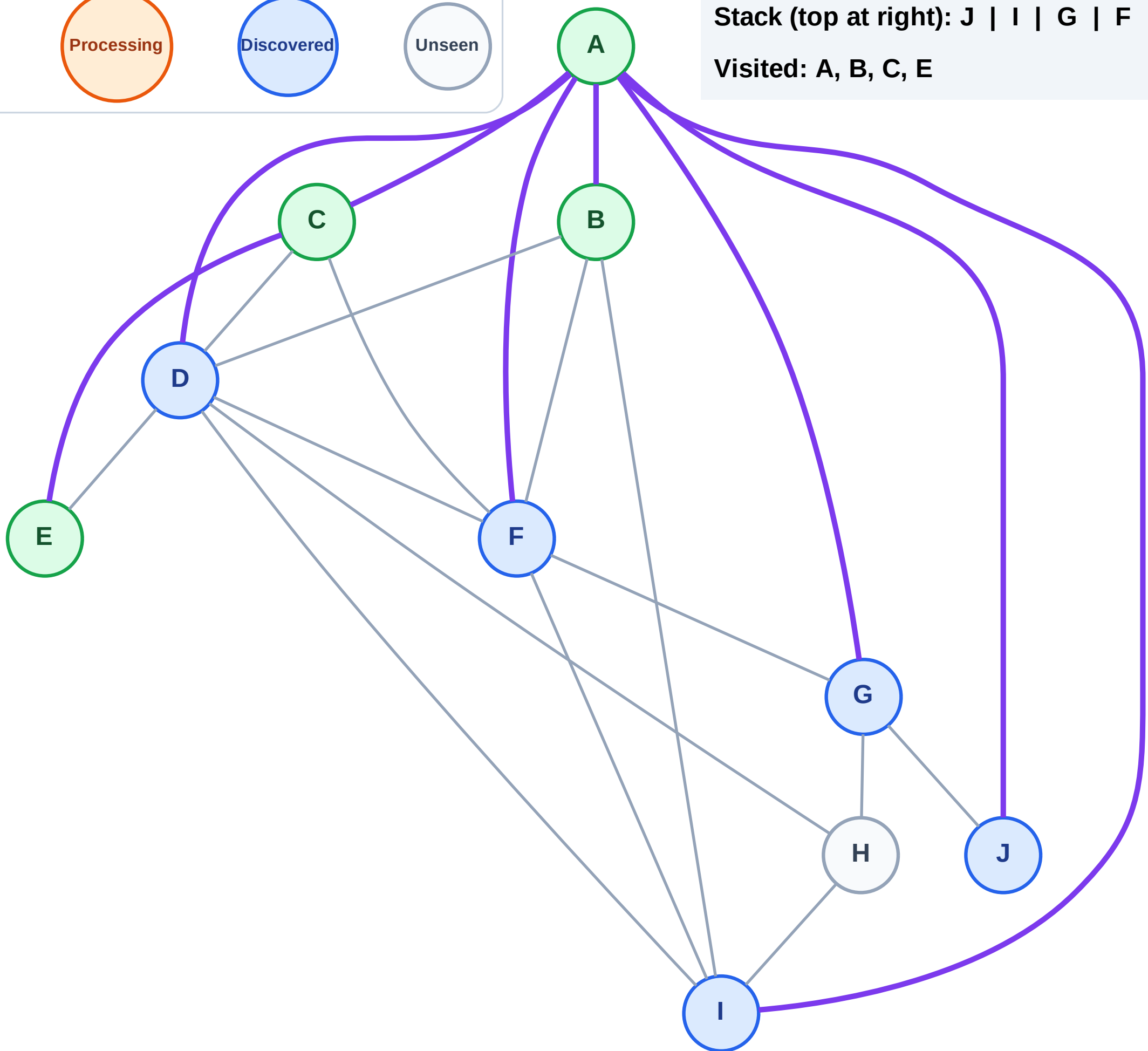
Step 9: No new neighbors from E

Legend



Stack (top at right): J | I | G | F | D

Visited: A, B, C, E



DFS Traversal

Step 10: Process node D

Legend



Complete



Processing



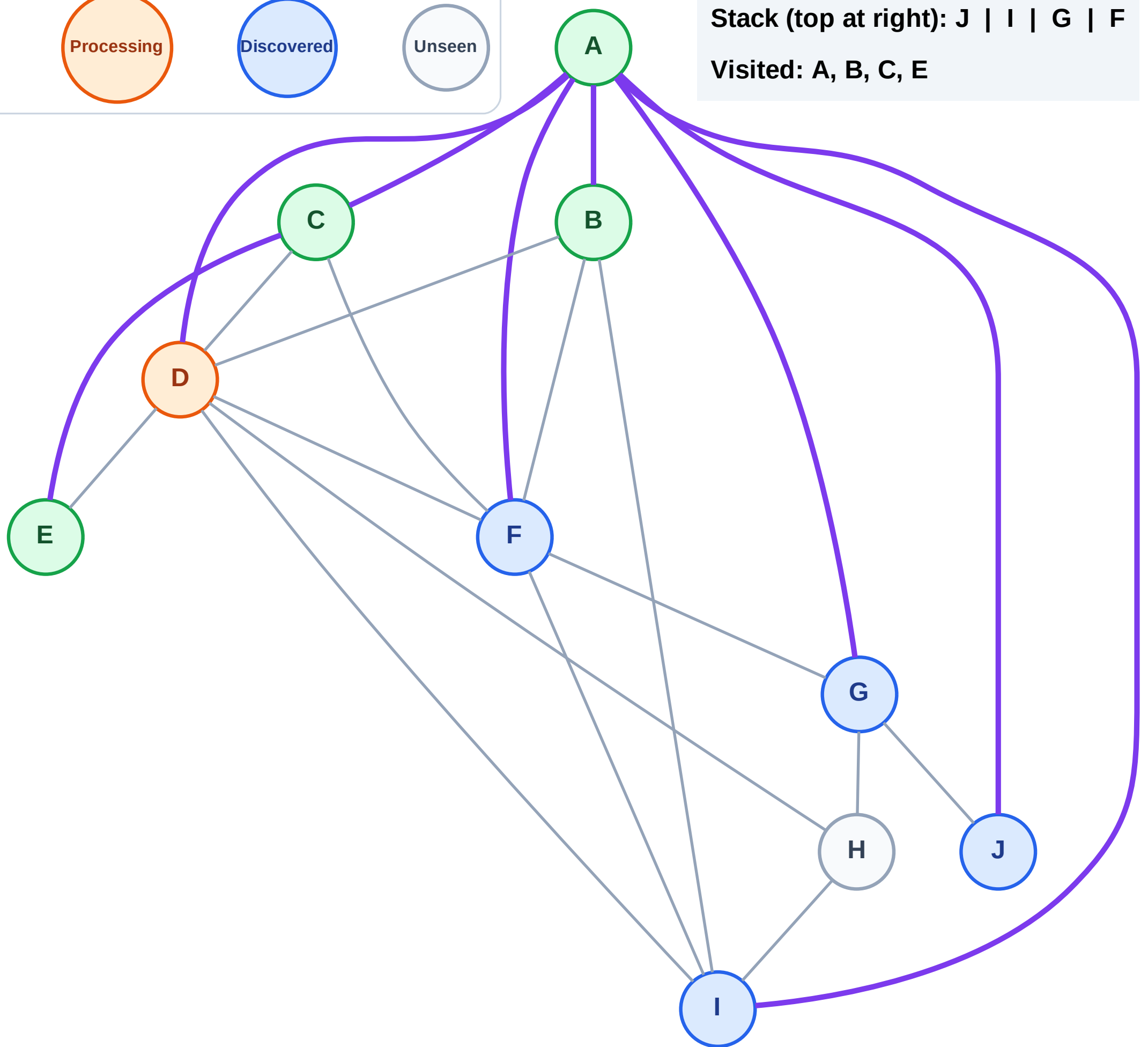
Discovered



Unseen

Stack (top at right): J | I | G | F

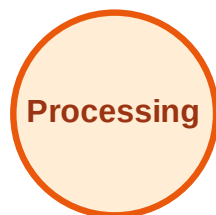
Visited: A, B, C, E



DFS Traversal

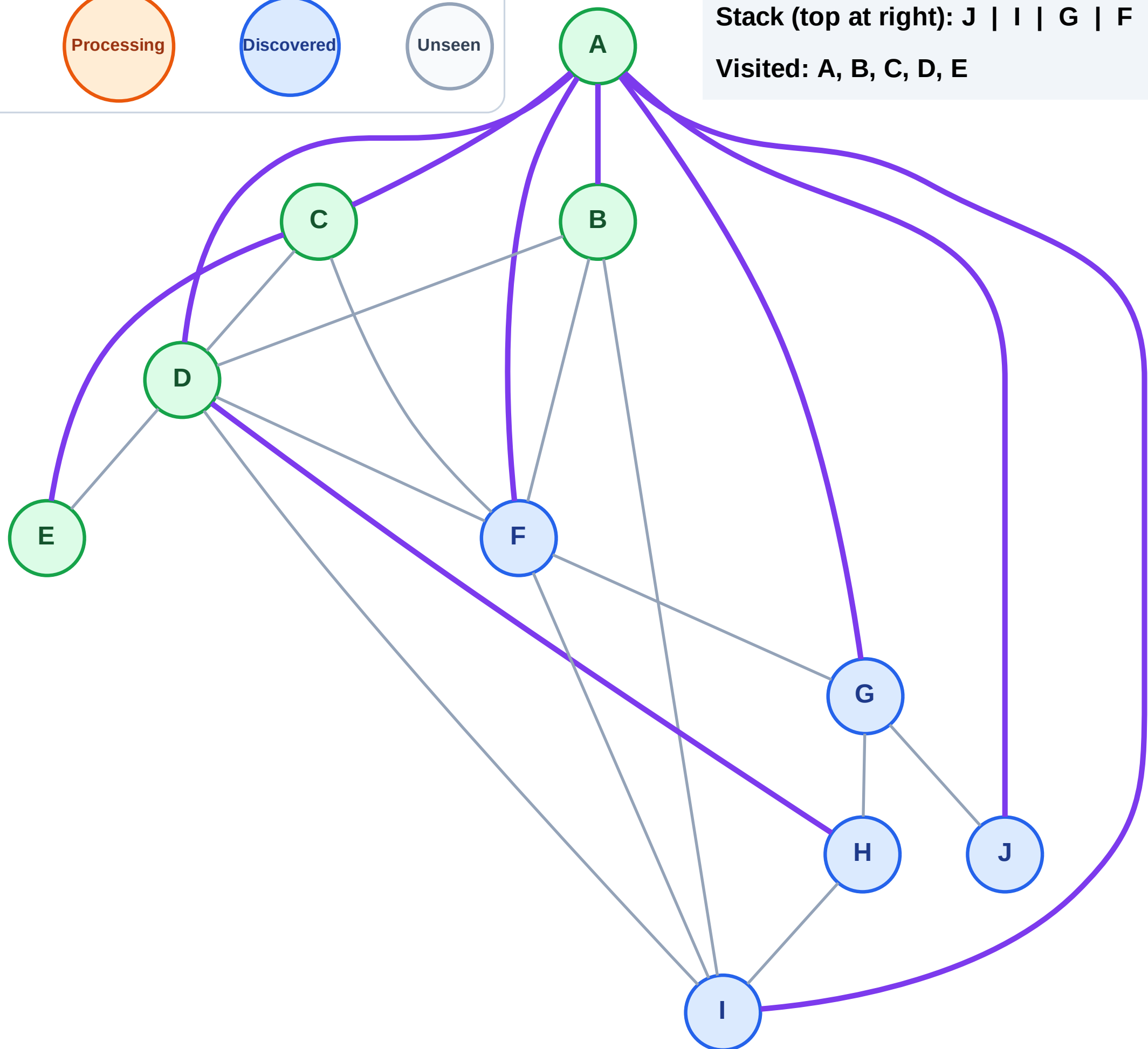
Step 11: Discover H from D

Legend



Stack (top at right): J | I | G | F | H

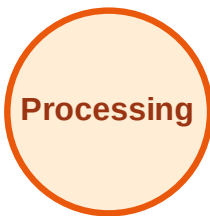
Visited: A, B, C, D, E



DFS Traversal

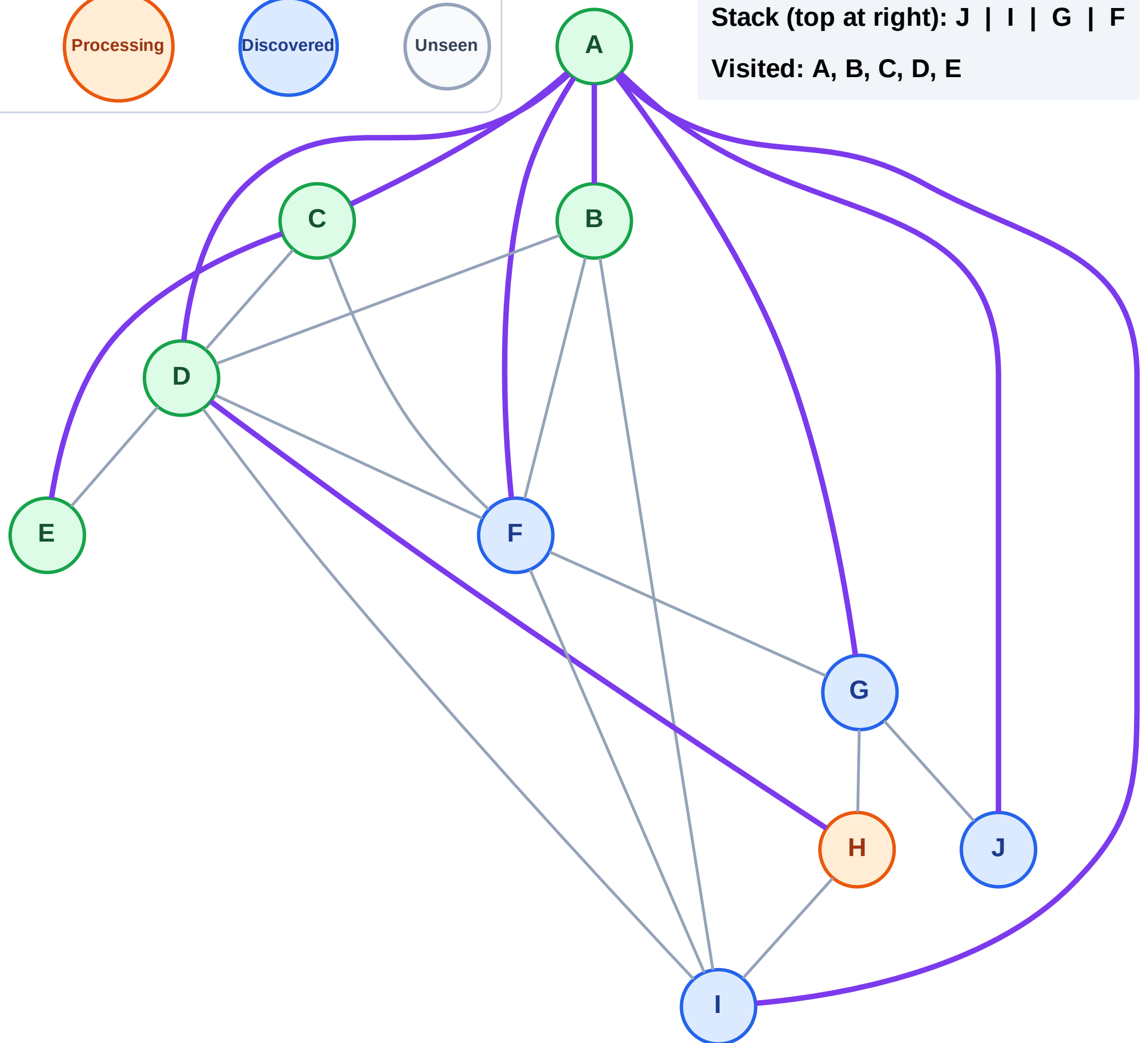
Step 12: Process node H

Legend



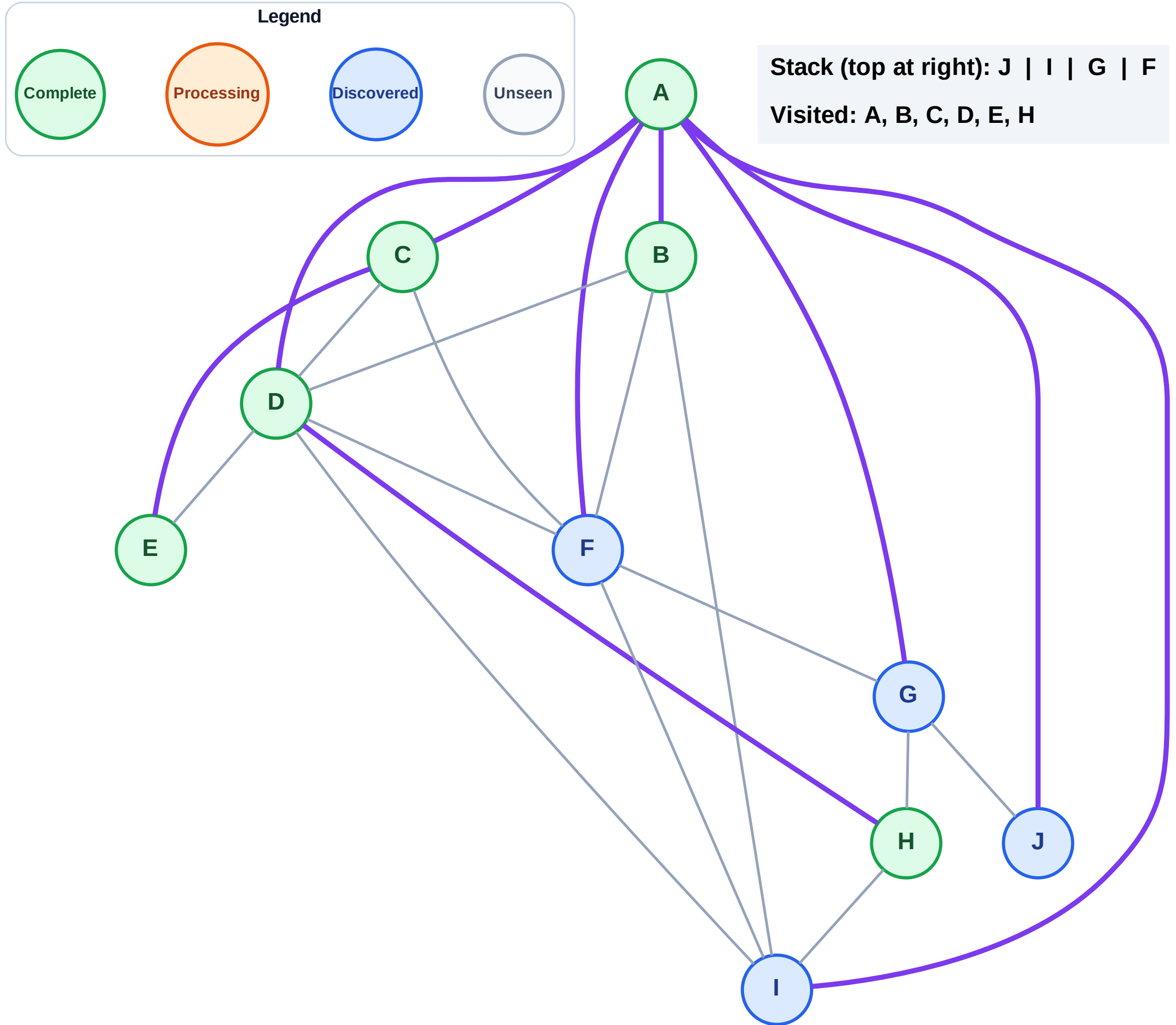
Stack (top at right): J | I | G | F

Visited: A, B, C, D, E



Step 13: No new neighbors from H

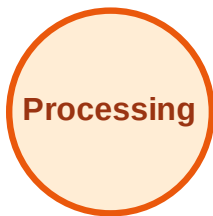
Step 13: No new neighbors from H



DFS Traversal

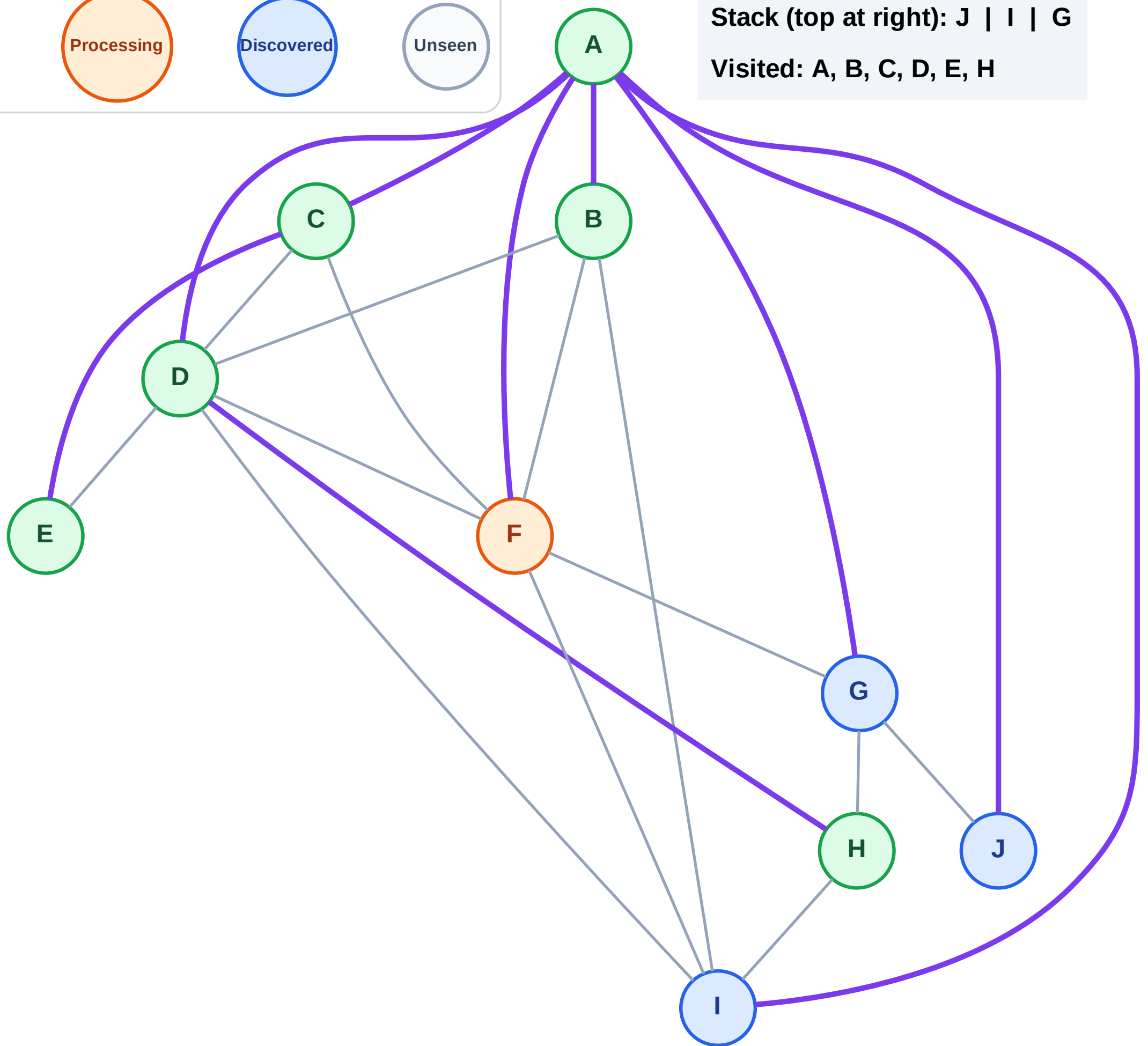
Step 14: Process node F

Legend



Stack (top at right): J | I | G

Visited: A, B, C, D, E, H



DFS Traversal

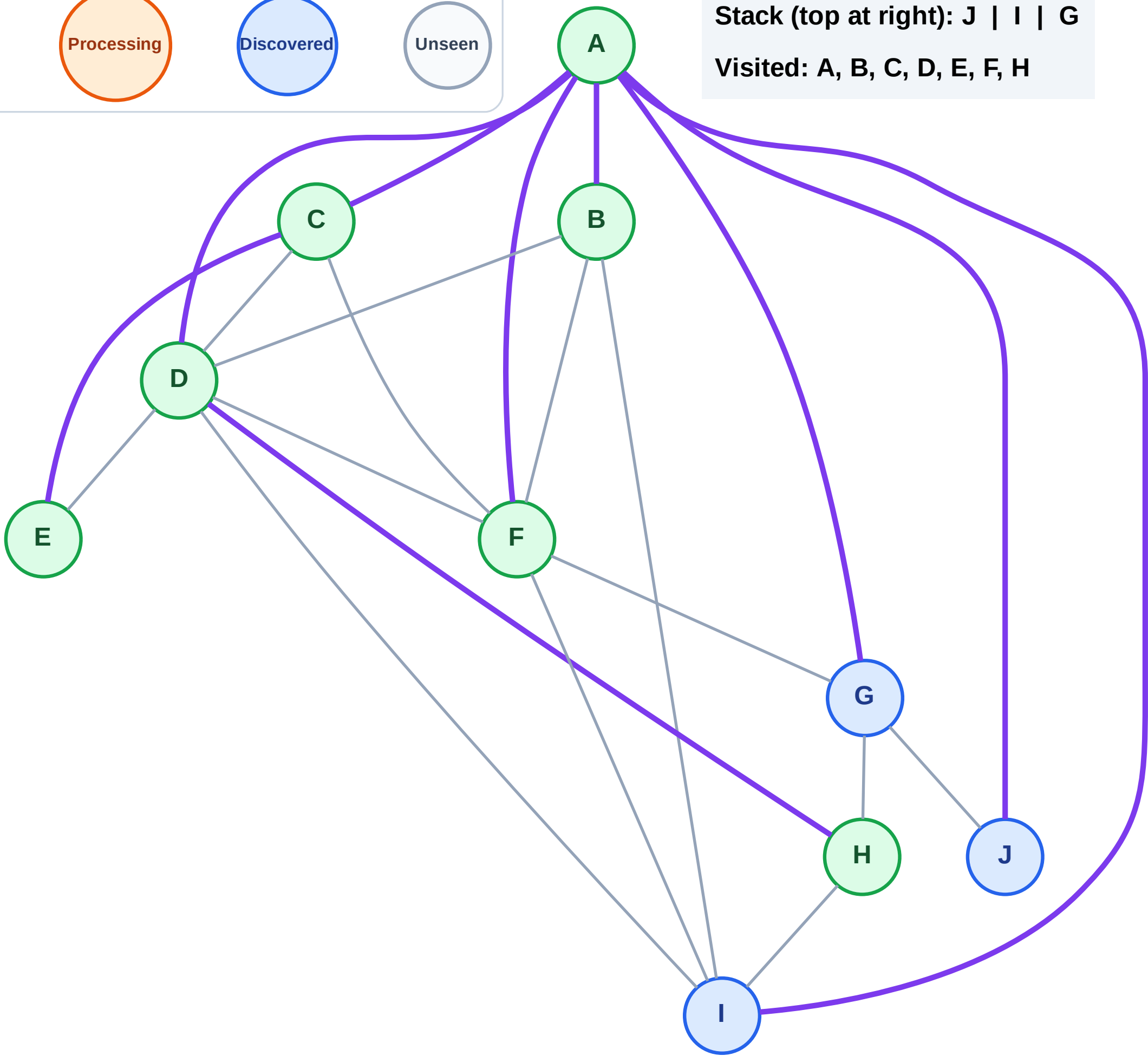
Step 15: No new neighbors from F

Legend



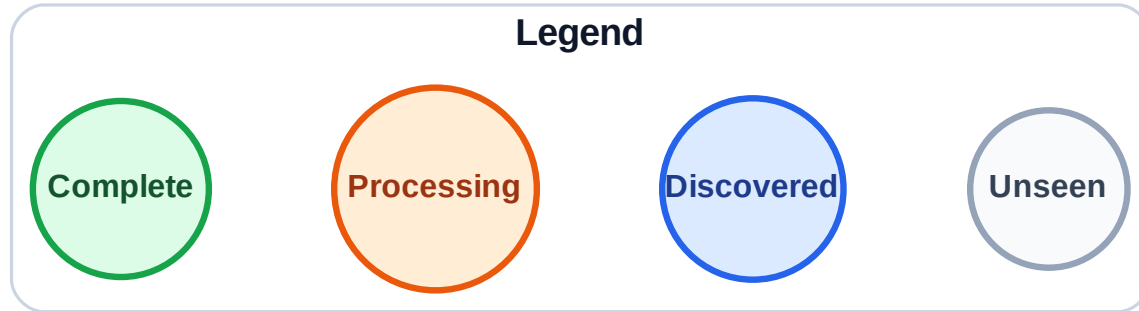
Stack (top at right): J | I | G

Visited: A, B, C, D, E, F, H



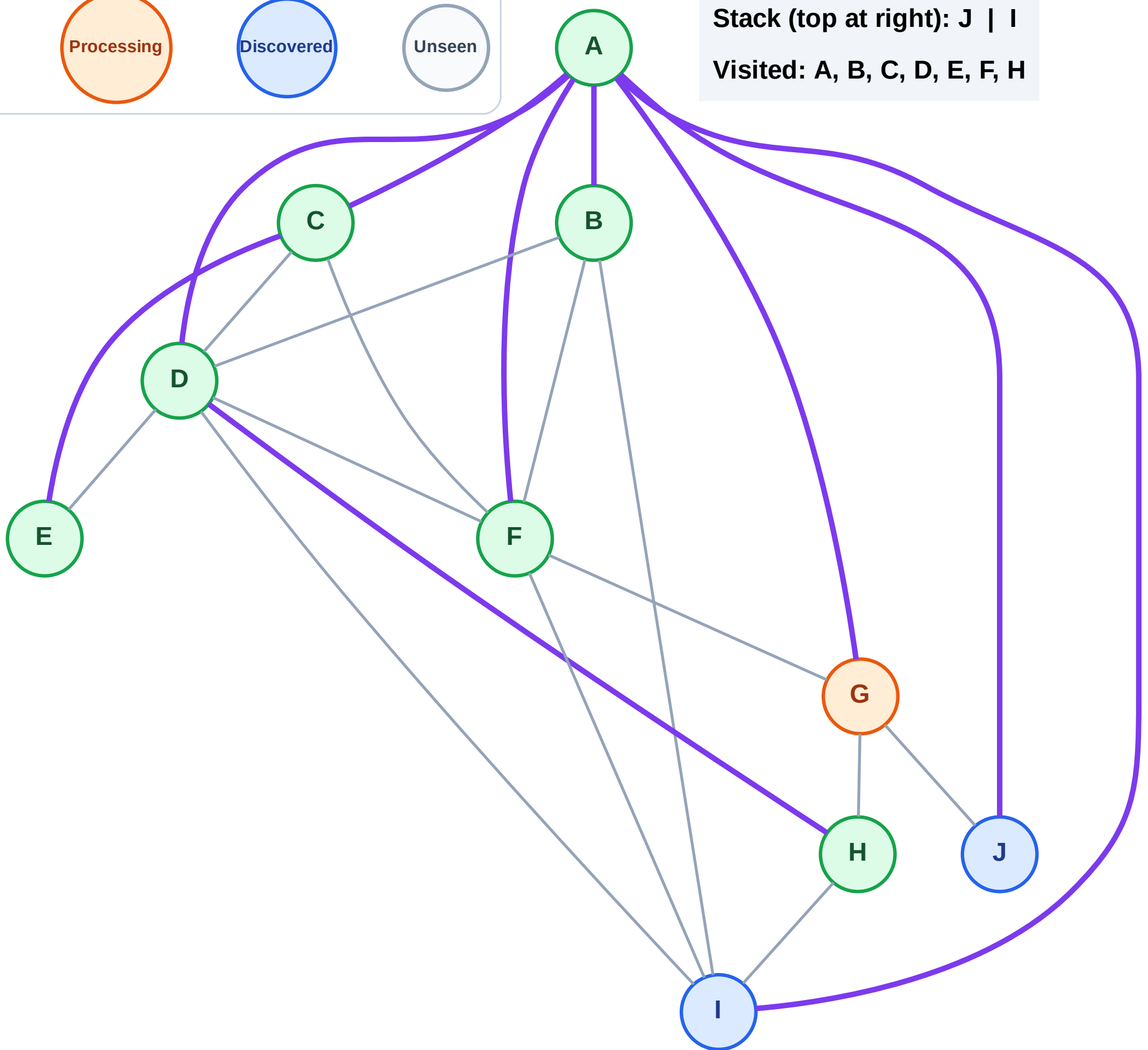
Step 16: Process node G

Step 16: Process node G



Stack (top at right): J | I

Visited: A, B, C, D, E, F, H



DFS Traversal

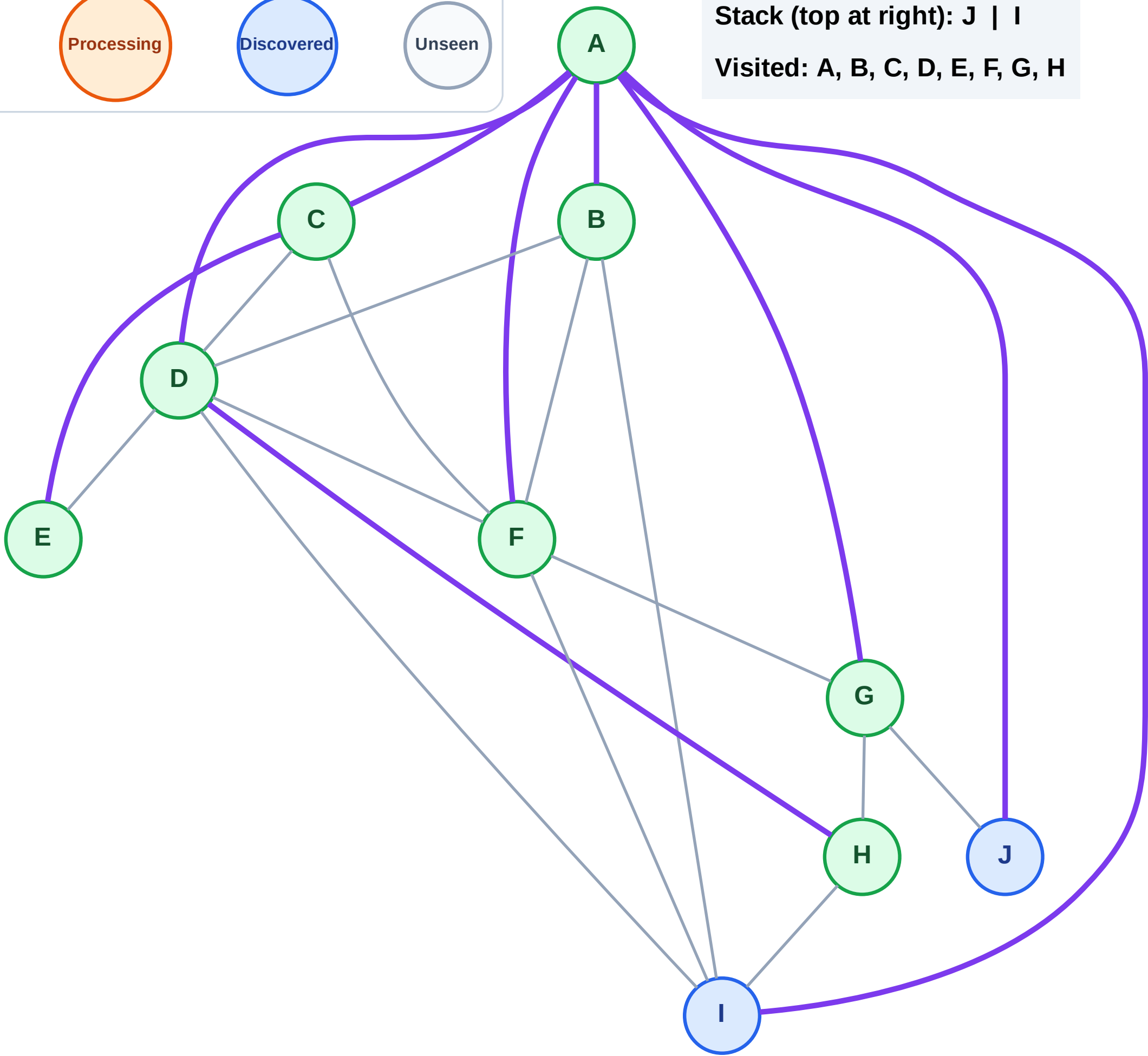
Step 17: No new neighbors from G

Legend



Stack (top at right): J | I

Visited: A, B, C, D, E, F, G, H



DFS Traversal

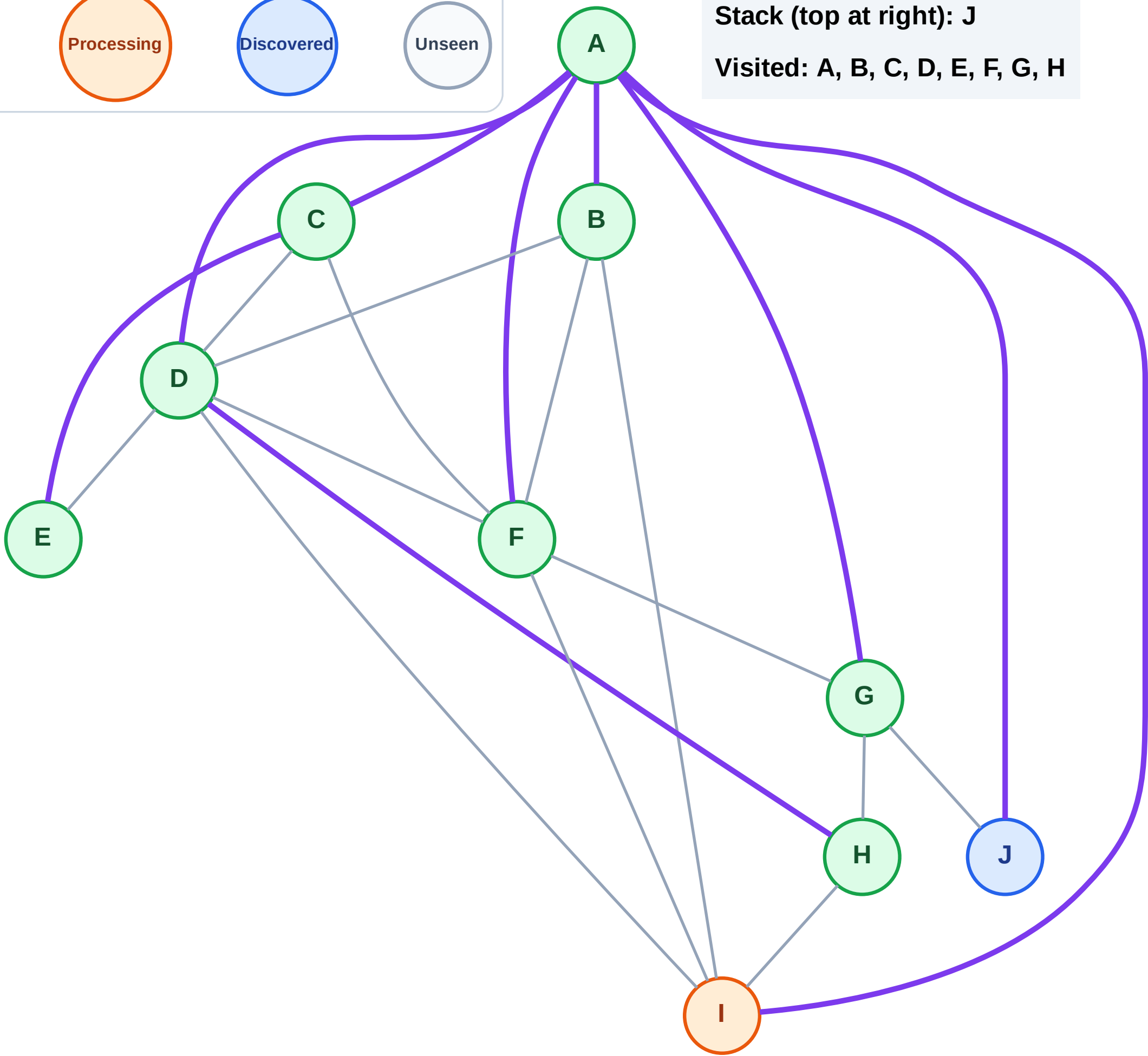
Step 18: Process node I

Legend



Stack (top at right): J

Visited: A, B, C, D, E, F, G, H



DFS Traversal

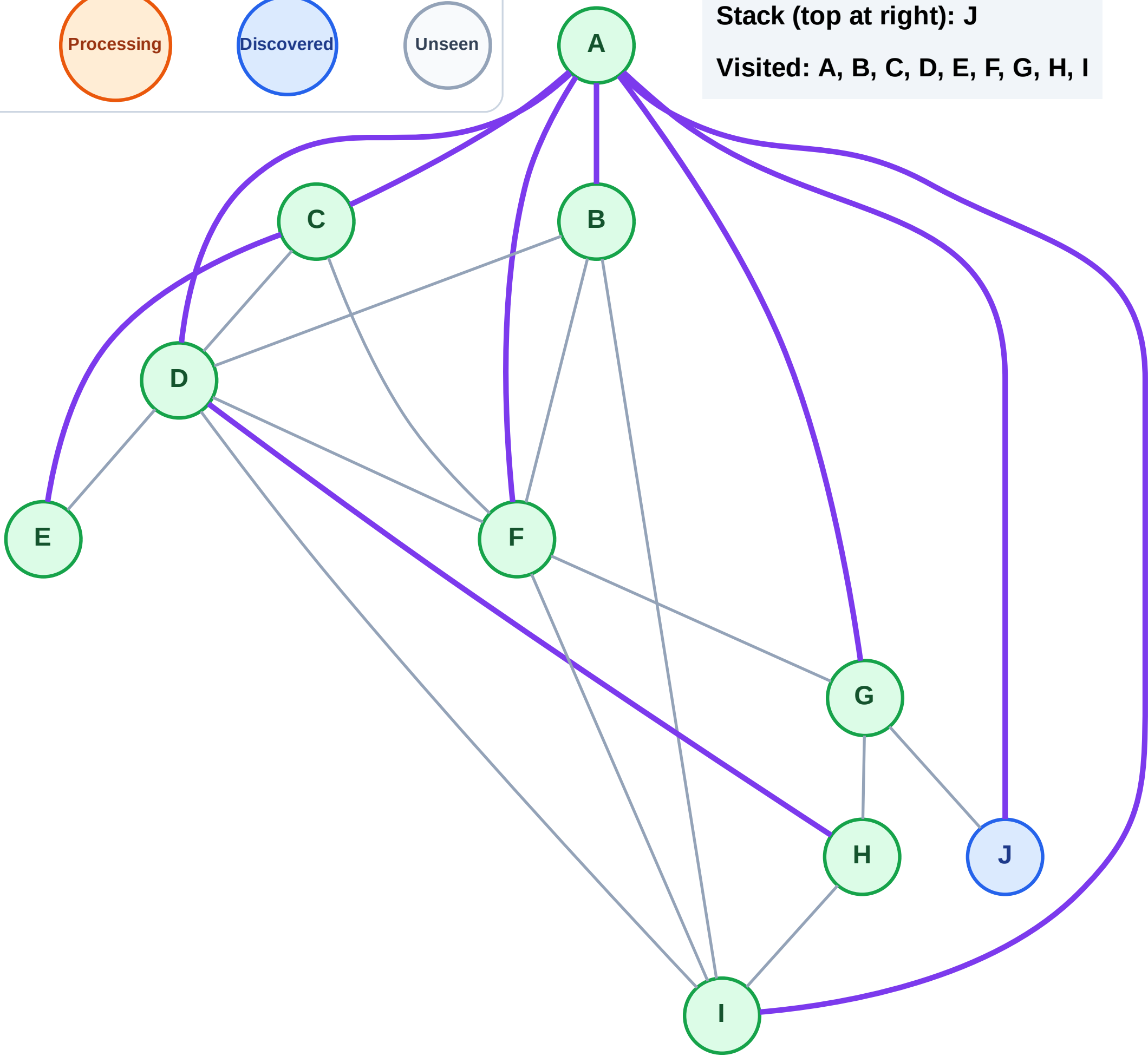
Step 19: No new neighbors from I

Legend



Stack (top at right): J

Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

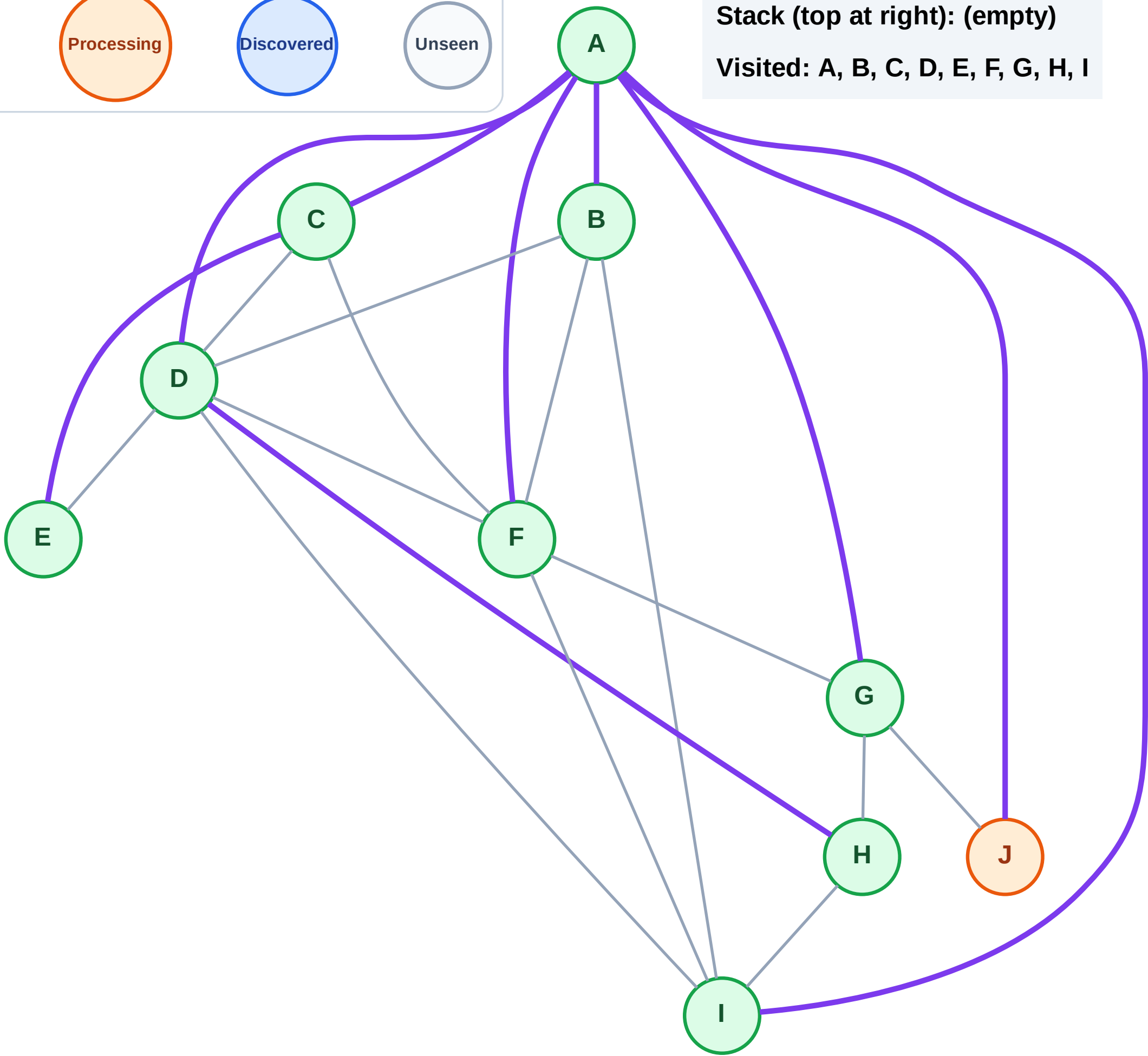
Step 20: Process node J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

